SECOND, COMPLETELY REVISED EDITION

# Handbook of Human-Computer Interaction

Editors: Martin G. Helander, Thomas K. Landauer and Prasad V. Prabhu

North-Holland

# HANDBOOK OF HUMAN-COMPUTER INTERACTION

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# HANDBOOK OF HUMAN-COMPUTER INTERACTION

## Second, Completely Revised Edition

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# PREFACE

The Handbook of Human-Computer Interaction is concerned with principles for design of the Human-Computer Interface. This is the second edition of the Handbook. The first edition was published in 1988.

The handbook has both academic and practical purposes. It is intended to summarize the research and provide recommendations for how the information can be used by designers of computer systems. There is an interest from academia in using this volume as a reference for teaching and research. The handbook can also be used by professionals who are involved in design of HCI, such as: Computer scientists, Cognitive scientists, Experimental psychologists, Human factors professionals, Interface designers, Systems engineers, Managers and Executives working with systems development.

Much information may be used outside the traditional field of HCI. With increased computerization we will find that there is a broad interest in HCI and usability. Many new users will not sit in front of a regular computer - for example users of smart consumer products, users of medical equipment, and car drivers who use automation in automobiles. Others will be interacting with very large computer applications, such as decision support in process industries and nuclear power plants, and software for product data management to name a few. Although we supply little direct information or research concerning these applications, the information in the handbook can be generalized to these areas. The main limiting factor in HCI is usually the cognitive limitations of the operator, and the implications of these for interface design. These are valid across all domains of application; to the extent that we can formulate theories of HCI these will be applicable in many areas.

Some readers may be interested in a definition of Human-Computer Interaction. We reluctantly supply a definition below. We are reluctant because it is generally difficult to capture a large investigative field with just a few words. Another problem is that there are a multitude of alternative definitions depending upon ones interest and goal. A professional interested in the organizational impact of software would have a different focus and framing of HCI than someone interested in, say, natural language understanding. Our general interest is to improve design of HCI., so our definition is also general.

*"In human-computer interaction knowledge of the capabilities and limitations of the human operator is used for the design of systems, software, tasks, tools, environments, and organizations. The purpose is generally to improve productivity while providing a safe, comfortable and satisfying experience for the operator".*

The contents of this edition of the handbook are fundamentally different compared to 1988. In the first edition there were 52 chapters. In this edition there are 62. Fourteen chapter titles remain since the first edition. Some of these have new authors and the remaining chapters have been rewritten or extensively modified to reflect the development in the field. The remaining 48 chapters are new. This reflects the development of the field: the continued search for new theories, new applications of HCI, and the fact that there has been a tremendous amount of new research since 1988. In the first issue of the handbook, there were some 5,000 references - now there are about 7,000, and most of them were published after 1988.

There are nine sections in the Handbook:

1. Issues, Theories, Models and Methods in HCI
2. Design and Development of Software Systems

3. User Interface Design
4. Evaluation of HCI
5. Individual Differences and Training
6. Multimedia, Video and Voice
7. Programming, Intelligent Interface Design, and Knowledge-Based Systems.
8. Input Devices and Design of Work Stations.
9. CSCW and Organizational Issues in HCI

Several of the topic areas that were hot issues in the first edition are not included or are less emphasized here. Examples of chapters that are no longer represented are: command names, text editors, and query languages. These topic areas are still of interest for HCI. However, the chapters in the 1988 edition provide authoritative reviews and they are still current. The problem in retaining them here is that most of the research was done before 1988, and little has been done after 1988. We recommend the reader of this text to consult the previous edition for such information. Below is a list of chapters from the first 1988 edition that are still current and deserve reading:

***Table 1. Chapters in the 1988 edition of the Handbook, that are still current and have become classics in the field.***

| |
|---|
| Chapter 1. D. D. Woods and E.M. Roth. Cognitive Systems Engineering |
| Chapter 2. J.N. Carroll and J.M. Olson. Mental Models in Human-Computer Interaction |
| Chapter 9. J. Rasmussen and L.P. Goodstein. Information Technology and Work. |
| Chapter 11. P.J. Barnard and J. Grudin. Command Names |
| Chapter 12. P. Reisner. Query Languages |
| Chapter 16. J. Elkerton. Online Aiding for Human-Computer Interaction |
| Chapter 20. H. L. Snyder. Image Quality |
| Chapter 24. D.E. Egan. Individual Differences in Human-Computer Interaction |
| Chapter 28. P. Wright. Issues in Content and Presentation in Document Design |
| Chapter 29. T.L. Roberts. Text Editors |
| Chapter 30. S.T. Dumais. Textual Information Retrieval |
| Chapter 36. J. Whiteside, J. Bennett. and K. Holtzblatt. Usability Engineering: Our Experience and Evolution. |
| Chapter 51. R.R. Mackie and C.D. Wylie. Factors Influencing Acceptance of Computer-Based Innovations. |

In this edition there are several missing areas, which would have been interesting to address, but they were not included because the chapters were either refused in the review process, or the authors did not deliver in time, or these areas were not included by the editors because there was not enough research. Here are some examples. Two chapter on the Internet were refused. Unfortunately, due to the tight time schedule in the production process they could not be replaced by other authors. Several chapters were not delivered by designated authors including a chapter on Display Image Quality. A potential chapter on HCI of software modules residing on the web was never considered, since this area has not been well researched. We therefore take exception to any potential comments by reviewers of this handbook that some important areas were not included. There were in most cases good reasons.

Each chapter was reviewed by two outside reviewers. The following individuals kindly contributed to the review process:

| | | |
|---|---|---|
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| David Travis | Gregg C. Vanderheiden | Robert A.Virzi |
| Pawan Vora | John Waterworth | Gunnela Westlander |
| Cathleen Wharton | Robert C. Williges | Dennis Wixon |

Several other individuals participated in the review process, and they are listed in the acknowledgments section of individual chapters.

We would like to acknowledge the contributions of the Department of Mechanical Engineering, Linköping Institute of Technology, Sweden, which paid for some of the travel expenses and telephone costs in coordinating the handbook. Thanks also to Dr. Anand Gramopadhye, of Clemson University, USA, for valuable help during the production editing of the handbook.

We are also grateful to Dr. Kees Michielsen of Elsevier Science. He found economic resources to organize the work on the handbook. He encouraged us throughout the production process and displayed great patience in waiting for the final product.

Most of all our thanks go to Davis Jebaraj, our production editor. He formatted and assembled the chapters, coordinated the subject index, and finally printed the text. He spent many long hours on this monumental task, and we owe him our sincere gratitude.

January, 1997

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# *Part I*

# Issues, Theories, Models and Methods in HCI

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*Handbook of Human-Computer Interaction*
*Second, completely revised edition*
*M. Helander, T.K. Landauer, P. Prabhu (eds.)*


# Chapter 1

# Human–Computer Interaction: Background and Issues

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One of us—we will not say which—writes these words with the help of a desktop computer that five years ago was as good as the industry had to offer, but that could be replaced today by a machine with 8 times as much RAM, 30 times as much hard disk capacity, 8 times the processing speed, and assorted additional capabilities (e.g., CD ROM, sound), at about 75 percent of its cost. The museum piece is retained because it is more than adequate to its user's needs; besides, why upgrade now when the machine that would be purchased is likely to be obsolete almost before it is unpacked? Information technology has been, and is, moving rapidly indeed.

The uses to which computers are put and the ease and effectiveness with which people interact with them have also changed over the past few years, but at nothing close to the same rate. Moreover, as we shall see, although many industries have invested heavily in computer-based tools for their workers, compelling evidence that the investment has paid off, either in terms of anticipated gains in productivity or increased satisfaction of most workers with their jobs, or most people with their lives, is wanting.

A major assumption that motivates this handbook and the work described in it is that the potential that information technology represents for helping people to attain their individual and corporate goals can be realized only if sufficient effort is made to discover and understand what determines whether a system will be useful and usable, or, better, how useful and usable a system will be. By information technology, we mean communication and computer technologies in combination.

Our purpose in this chapter is to help set the stage for what is to follow by discussing, briefly and in broad terms, the history, state, and prospects of information technology, to consider a representative sample of HCI issues relating to the uses of this technology, and to raise some questions for future research, innovation and development. The focus of the handbook is on people interacting with—using—computers, and information technology more generally, so there is relatively little emphasis on hardware per se; our intent is to say only enough to provide a sketchy account of the technological context in which the issues of interaction arise.

## 1.1 Historical Background

### 1.1.1 Communication Technology

Until late in the 19th century the speed of long-distance communication was limited by methods of transportation. To transmit a message from point A to point B, one put it on some physical medium, and transported that medium from the one point to the other. Various methods were devised to obviate the transporting of messages, including the use of smoke signals, flashing lights, or sounds that could be sent and resent through a series of appropriately spaced relay stations. Such systems had severely limited message-carrying capacity and could be effective only over relatively short distances.

The inventions of the telegraph and the radio constituted major advances in communication technology. Both made it possible to send messages over very long distances extremely rapidly. The telegraph, and later the telephone, became the means of choice to effect long-distance communication between individuals. The radio, as it evolved, and later television, were used predominantly for broadcasting. As a matter of historical interest, when radio was first used, it was seen primarily as a means of point-to-point communication; only later did its broadcast possibilities become widely recognized and this mode of operation become the dominant one; Alexander Graham Bell and his early supporters thought the telephone would be primarily a broadcast medium, sending musical performances far and wide. Now, with the rapid spread of cellular (radio) telephony, and cable connections with their potential for interactive use, the association of media with role and content is again in the process of change.

### 1.1.2 Computer Technology

Precursors to modern computer technology can be seen in ancient devices used to facilitate computation (the abacus or soroban) as well as in less ancient inventions (Pascal's 17th century mechanical calculator, Jacquard's 18th century loom, Babbage's 19th century analytic engine, Hollerith's tabulating machinery used in the census of 1890), but the beginning of the age of the electronic digital computer is probably best marked at roughly the middle of the 20th century. One might debate which of several events most deserves to be considered the starting point; among the possibilities would be the completion of Howard Aiken's Mark I by IBM in 1944, that of J. Presper Eckert and John Mauchly's ENIAC with its 17,000 vacuum tubes in 1946, the invention of the transistor by John Bardeen, Walter Brattain and William Shockley in 1948, and the first commercial sale of an electronic digital computer—a Univac—in 1951.

The important point, for present purposes, is that the electronic digital computer is only about half a century old. Despite this fact, it has already had an enormous impact on our lives, and the technology is still in a phase of rapid development; what its further impact will be is unclear, but who can doubt that it will be very great? The speed with which computer technology has advanced since the middle of the century is remarkable by any standards, and "predictable" only with the benefit of hindsight. Few, if any, of even the most clear-sighted visionaries of fifty years ago imagined in their wildest speculations how things would develop over the next few decades.

The most apparent trend that has characterized progress in the technology over that time has been the ever-decreasing size of computing components ("active element groups," such as logic gates and memory cells), which has allowed the packaging of ever-larger amounts of computing power into less and less space. With the miniaturization have come also great increases in the speed and reliability with which components operate, and commensurate decreases in their costs and power requirements. The aggregate effect has been to make much greater amounts of computing power than were available to major corporations even 20 years ago affordable today to individuals of modest means

The surprising nature of what has transpired may be illustrated by the juxtaposition of two observations published in the Scientific American, one in 1949 and the other in 1994. In 1949 George Gray, citing Warren McCulloch as the source of his estimates, noted that "if a calculator were built fully to simulate the nerve connections of the human brain, it would require a skyscraper to house it, the power of Niagara Falls to run it, and all the water of Niagara to cool it." Forty-five years later, Marvin Minsky (1994), noting that the brain is now believed to contain on the order of 100 trillion synapses, speculated that "[s]omeday, using nanotechnology, it should be feasible to build that much storage space into a package as small as a pea" (p. 112). Minsky's speculation is admittedly a little noncommittal as to when this is likely to be feasible, but even a claim that it would ever be so would have seemed absurd in 1949.

Other indications of the spectacular rate of change in computer technology are not difficult to identify. The cost of a logic gate went from about $10 in 1960 to about $0.0001 in 1990. The estimated number of active element groups in the average U.S. home went from about 10 in 1940 to about 100 in 1960 to about half a million in 1990.

### 1.1.3 Software

From the beginning, the development of software has been at least as great a challenge as the continual improvement of hardware, but progress has been more sporadic and more difficult to judge objectively. In the early days of computers, most users were programmers as a matter of necessity. Prewritten programs did not exist in abundance, so if one wanted to use a computer to accomplish some task, one had to write the required program oneself. Today, commercially available applications programs abound.

The nature of programming has changed more or less continuously since computers first appeared on the scene and had to be programmed in machine code. The variety of languages and operating systems that exist make it difficult to say much about programming that is applicable in all contexts. At the highest level of abstraction, programming can be thought of as a two-phase process: (1) specifying in detail the steps that are necessary to accomplish some task, and (2) expressing those steps in a language of a particular operating system. In practice, the two phases may be so closely coupled that they cannot be distinguished, but they are conceptually different; the same procedure specification may be represented in different languages so as to be executable on different systems.

Programming today can be done at a variety of levels. Systems programmers develop operating systems like DOS, Windows and UNIX that represent operating environments in which other programs can run. Applications programmers produce programs to be used for specific purposes, like maintaining payroll information, scheduling airline travel, composing and editing text, doing mathematics, helping to diagnose illness, and designing industrial products. Starting with the invention and dramatic evolution of compilers, programming languages have evolved in several ways: by taking over more of the chores originally done by programmers, such as translating abstract symbols, words, control instructions and operators into detailed sequences of basic operations by which they are achieved; by providing bookkeeping and organizational aids, such as automatic memory allocation, configuration management, and version control; and by enabling and enforcing various notational schemes, systems for expressing logical relations or procedures, and disciplines for the expression of relations and ideas, such as are represented by LISP and object oriented languages.

As a result of these developments, much bigger, more complicated programs became easier to construct, and the rate at which new software systems could be developed increased greatly. Now, even relatively unsophisticated computer users can write their own programs, if they wish, using specific languages provided with most desktop or laptop computers for that purpose; however, probably only a small minority of personal computer users actually exercise this option. Basic HCI efforts such as user-centered design and iterative evaluation have played little role in the development of programming languages, although such efforts might be richly rewarded, especially, perhaps, in the case of the expanding availability of programming facilities for nonprofessional users.

### 1.1.4 The Blending of Communication and Computer Technologies

Information technology, as the term is generally used today, and as it is used here, connotes a blending of communication and computer technologies that has been so thorough that it is no longer possible to tell where the one ends and the other begins. The most obvious and significant result of this blending has been the building of computer networks, beginning only a little more than a quarter of a century ago, that link computers from different geographical locations around the world (Cerf, 1991).

Computer networks combine computer and communication technologies in the obvious sense that they connect computers through communication channels, but the blending of the technologies goes much deeper than that. The communication networks are themselves permeated with computational capabilities that are essential to their operation. The blurring of the distinction between computer and communication technologies is seen not only in the implementation of computer networks, per se, but in many of the capabilities and services that such networks provide. These include electronic mail and bulletin boards, computer-mediated teleconferencing, multifeature phones and voice mail, facsimile store-and-forward capabilities, digital encryption systems, information utilities, and many others.

### 1.1.5 Worldwide Connectivity

The first computer networks were implemented on an experimental basis in the mid 1960s (Davies and Barber, 1973; Marrill and Roberts, 1966). The ARPAnet, the immediate predecessor of the Internet, began in 1969 as a four-node system funded by the Advanced Research Projects Agency of the U. S. Department of Defense in 1969 (Heart, 1975; Heart, McKenzie, McQuillan and Walden, 1978); it became and remained the largest operational network in the world for many years. Its successor, the Internet connected an estimated 1.7 million host computers and had an estimated 5 to 15 million users as of 1993 (Pool, 1993). The 1995 special issue of Time on Cyberspace reported 4.8 million hosts (Elmer-DeWitt, 1995).

Traffic on the NSFnet backbone of the Internet increased 100-fold between 1988 and 1994, and as of 1994, traffic on all federally funded networks and the number of new local and regional networks connecting to them was doubling annually (OSTP, 1994). Of course all of these numbers are estimates, and any of them could be in error. Even allowing for some inaccuracies, however, it seems safe to assume that the community of users of computer networks is growing very rapidly and is likely to continue to do so for the foreseeable future.

China got its first direct link to the Internet in May, 1994; it was used to support 2000 terminals at the Chinese Academy of Sciences and two of its major universities. Two more links were scheduled to be established early in 1995 (Plafker, 1995). These are thought-provoking events both because China is the most populous country on the planet and because, as a society, it has, for a long time, been relatively isolated from much of the rest of the world. One can only guess what the effects will be—on China and the rest of the world—of the international communication that easy access to the global networks would make possible. More generally, what the eventual implications (psychological, social, economic, political) will be of the kind of worldwide connectivity we anticipate is very difficult to say. The one point on which technologists clearly agree is that they are very likely to be profound (Hiltz and Turoff, 1978; Licklider and Vezza, 1978). For HCI, the most obvious implication is that the number and diversity of users of computers is likely to continue to grow rapidly over the period during which this book will serve its audience. And the majority of the vast numbers of new users are likely to be less sophisticated than those of the past, at least for the next several years, if not permanently. The challenge to HCI is apparent.

How long it will take is anybody's guess, but it seems likely that eventually information will be moved around from place to place by an integrated technology that will incorporate the capabilities now provided by the telephone, radio and television, and computer terminals that are linked to networks. With the same device one will be able to send and receive text, voice and video, and to do so to and from other people as well as to and from information resources and services of many types. Ensuring that the functions and interfaces for that system are useful and usable, given the enormous heterogeneity of the user population and its goals, will be a major task for the HCI community.

## 1.2 Expectations

### 1.2.1 Basic Technology

Electronic computing and storage devices can be improved beyond present capabilities—in terms of switching speed and storage capacity—before fundamental limits to progress are met. Further increases in speed are being sought through the making of transistors of materials other than silicon, such as gallium arsenide and alloys of silicon and germanium (Meyerson, 1994) as well as through further reductions in scale.

Memory chips with a billion bits of storage capacity and microprocessors capable of performing more than a billion instructions per second are currently being designed. This is a 1000-fold increase over the technology of the early 1980s in both cases (Stix, 1995). The smallest dimensions on the most recent memory chips are about 0.35 micron and are expected to reach 0.2 if not 0.1 micron ("point one") by early in the coming decade. At about .05 microns, dimensions are approaching those of some biological molecules and devices are beginning to be subject to disruptive quantum effects, so miniaturization probably cannot advance much beyond this point without a shift to new principles of operation.

The ability to store very large amounts of information in very small spaces is, of course, already being exploited for the purpose of making information in encyclopedic quantities more readily available to specialists and general consumers alike. *The Journal of the American Chemical Society* was made available on CD ROM in 1994. "The 1994 edition of Microsoft Bookshelf contained on one searchable CD ROM the complete text of the *American Heritage Dictionary, Roget's Thesaurus*, the Columbia *Dictionary of Quotations*, the *Concise Columbia Encyclopedia* (15,000 entries and 1,300 images), the *Hammond Intermediate World Atlas* and the *World Almanac and Book of Facts 1994*—not to mention tricky pronunciations of foreign places in digital stereo" (Eisenberg, 1995, p. 104).

One vision of what will be available to the individual computer user by the end of the century or soon thereafter includes a battery operated 1000-by-800-pixel high-contrast display, a fraction of a centimeter thick and weighing about 100 grams, driven by a low-power consumption microprocessor that can execute a billion operations per second and that contains 16 megabytes of primary memory, augmented by matchbook size auxiliary storage devices that hold 60 megabytes each, and larger disk memories with multiple gigabyte or even terabyte capacity (Weiser, 1991).

As this technology based on electronics continues to advance, alternative types of devices are being researched that promise eventually to go beyond what electronics can deliver. Devices that use photonics instead of electronics, for example, have existed in experimental forms for several years and may be available commercially before many more, and ones based on biological molecules may not be too far behind (Birge, 1994, 1995; Horgan and Patton, 1994; Parthenopoulos and Rentzepis, 1989). One implication of these predictions is that HCI design may continue for the foreseeable future to be predicated on successive waves of extraordinary changes in the capacities of the underlying technologies that support our inventions and designs.

Most of the speculation about the nature of human–computer interaction in the future is based on the model of a person using a device that obviously is a computer. Although this model is likely to continue to be appropriate for much of human–computer interaction, at least in the near-term future, it is also certain that in many instances the computers with which people will interact in the future will not resemble computers as we think of them today. They will be invisible—embedded in the woodwork, as it were. As a direct consequence of the spectacular success that designers have had in packaging enormous amounts of computing capability in tiny devices that can be manufactured at very little cost, computational capability is being incorporated not only in "large ticket" items like automobiles and major appliances, but in furniture, games, toys and clothing. This increasing distribution of computing capability is likely to have implications for the study of human–computer interaction that we have not yet imagined.

### 1.2.2 Software Development

Software continues to be a major challenge. According to Gibbs (1994), underestimation of the difficulty (as measured by time and money) of developing new systems and bringing them to the point of operation is a chronic problem. "Studies have shown that for every six new large-scale software systems that are put into operation, two others are canceled. The average software development project overshoots its schedule by half; larger projects generally do worse" (p. 86). Gibbs describes several very large and highly visible software development projects the costs of which escalated to several times the original estimates before the projects were terminated or forcefully scaled down. Although development managers sometimes believe that adding usability engineering activities and user testing to projects will cost even more, a case can be made that the opposite effect, saving time by simplifying requirements and avoiding costly revisions, is a more common result (see Landauer, 1995, p 320–322)

Programs containing millions of lines of code have become commonplace. Such programs are sufficiently complex that no individual understands them in their entirety. Guaranteeing that they are error-free, or that whatever errors they contain will not lead to catastrophic malfunctions, is very difficult—perhaps impossible. Techniques are needed that will provide reasonable estimates of probabilities of malfunctions and of their likely consequences. Quite good methods exist for estimating interface flaws (Landauer,1995, p 310–313). Research into (programmer-) user-centered methods and tools for *preventing* programming errors is also needed.

Despite the difficulties involved in software development, a lot of useful software is being produced, much of which is readily available to anyone who wishes to acquire it. This is not to suggest that all, or even most of, the software that gets to the commercial market is useful, or that it delivers what its promotional literature claims, but some of it is, and does, and more is being developed all the time. However, as we will elaborate below, the greatest barrier to usefulness has been too little effective attention to how, for what purposes, and with what results the technology is being put to use.

### 1.2.3 Users and Uses

When computers first appeared on the scene, users were not, for the most part, garden-variety folk. They tended to be technically trained, or at least technically oriented, and to find programming to be an intrinsically interesting activity. Many were willing to invest large amounts of time in learning what was necessary to use these machines and to endure all sorts of inconveniences to get a modicum of the machines' time to execute their programs.

Although computer professionals and hobbyists still form a large segment of the most frequent users of computers and networks, today's computer users include people from all walks of life, many of whom are neither technically trained nor even interested in technology per se. They use computers not for the purpose of developing the technology, but as a means to other ends: document preparation and editing, accounting and bookkeeping, information access, decision aiding, music composition, designing (buildings, industrial equipment, consumer products), and for countless other purposes. People who use computers more or less daily in their work include, sales people, scientists, hotel clerks, engineers, librarians, gas station attendants, secretaries, stock brokers, airline reservation personnel, bankers, physicians, supermarket cashiers, law enforcement officers—it might be easier to list people who do not make use of computers.

Computer technology now pervades the workplace, and its applications here continue to increase rapidly (Sproull and Kiesler, 1991a, b). Many jobs have changed significantly as a consequence of its introduction; some have been transformed beyond recognition. In many cases, the workplace itself has been relocated to the home or, in effect, to wherever one cares to take a portable computer. According to Jaroff (1995), three million employees of American companies were telecommuting, at least part of the time, for purposes of work as of 1995, and this number was increasing by about 20 percent per year. The transient and long-term implications—economic, psychological, social, technical and for HCI—of this change remain to be understood.

Computers are used not only for work, but for avocational and educational purposes, for entertainment, for communication, for personal information management, and, unfortunately, for crime and various types of mayhem as well. Users include very smart people and less smart people, people of all ages, people who are able bodied and people who are disabled in one or another way, people of all ethnic origins and every conceivable political persuasion, urbanites, suburbanites, and people from relatively isolated parts of the world; in short, it is hard to imagine a more heterogeneous community than that constituted by computer users today. Indeed, the user community that has to be considered is already so heterogeneous and expanding so rapidly that, for design purposes, we would not go far wrong in assuming that it includes essentially everyone.

One of the major surprises to follow on the implementation of government and commercial computer networks was the degree to which they were used for purposes of person-to-person communication via electronic mail and associated resources. This was not the use that was intended or expected by the original network implementors; nevertheless within two years of the installation of the ARPAnet, electronic mail was its major source of traffic (Denning, 1989) and, so far as we know, it (along with the closely associated electronic bulletin boards and forums) remains the primary use today, at least if usage is measured in terms of numbers of users or occasions of use, rather than in terms of amount of computing resources—storage and processing cycles—utilized. Whether person-to-person communication will continue to be the most common reason for using computer networks, there can be little doubt that it will remain among the more common uses.

Steinberg (1994) identifies, as the four major uses of the Internet, email, electronic bulletin boards, real-time conversations, and retrieval of information from electronic databases and libraries. Real-time conversations take place within IRCs (Internet Relay Chats) or MUDs (Multi-User Dimensions) which link people who are on line at the same time and, in the case of MUDs, provide them with virtual places or environments (virtual rooms and other imaginary places) that are thematically suited to the real-time interaction that takes place. IRCs and MUDs appear to be used primarily for casual conversation. More generally, Steinberg claims that "most people who inhabit the Net use it chiefly for human-to-human contact, not for gathering information from computer databases" (p. 29). If, as Steinberg surmises, a major reason for this is the fact that information on the net is hard to find, this situation might change with the development of more effective information finding aids, of which programs like Gopher, Archie, Mosaic, Netscape and others may be considered promissory notes awaiting, at least in part, advances based on HCI research and on more attention to user-centered design and evaluation.

In the most recent explosion of network usage, graphic user interfaces (GUIs) and associated search software, formatting, and message-passing standards have spawned an enormous interest in the World Wide Web (WWW), a method for connecting computer networks everywhere. Individuals, institutions and commercial organizations rushed to establish presence on the national and international network scene by creating "home pages" for themselves and stocking them with information and decorations to appeal to others. Network-based ventures such as flower shops, virtual vineyards, movie recommenders, telephone service desks and email order catalogue sales are, as of this writing, sprouting like mushrooms after rain. New interfaces, browsers, search engines and other facilities and accessories are becoming available at a bewildering rate.

The topsy-turvy race to create these new resources is at once impressive and daunting. While there are ever more marvelously large and heterogeneous information sources, services and facilities to choose among, there is an ever more difficult problem of search and choice. This issue of making the networks provide ease and quality to individual users, as compared to sheer quantity of access, may be the single greatest challenge facing HCI in the near future.

### 1.2.4 Tools and Services

As the users of computers have grown in number and the uses to which computers are put have become increasingly diverse, the number and diversity of tools and services that the industry has to offer have increased apace. All systems and software packages come with users' guides and handbooks; many, if not most, also offer on line instruction and tutorial aids. The usability of this material varies considerably from system to system, but our impression is that it is not easy, as a general rule, to become an adept user of any nontrivial system simply by consulting instructional material, whether packaged in book form or presented interactively on line. The proliferation of third-party how-to books, courses and consulting services to help users of popular applications programs is evidence of this problem. (Perhaps this fact should be taken as a challenge for HCI; after all, "how-to" books are not in great demand for users of telephones, cars, or household appliances.) Learning from experienced users appears to remain a favored and relatively effective approach.

### 1.2.5 Increasing General Awareness of Information Technology

Human–computer interaction is not a subject that one was likely to have learned anything about through the mass media until very recently. Within the last few years, however, this has begun to change. Significant amounts of space are now being devoted to the topic by major news magazines and periodicals; it is not uncommon for some aspect of the topic to be a cover story, and occasionally it accounts for an entire issue, to wit the Spring, 1995, special issue of Time on Cyberspace. The topic is even receiving explicit recognition in political agendas.

Many terms that would have been considered esoteric technical jargon a short time ago—RAM, modem, cyberspace, email, Internet—have made their way into common usage. Elmer-DeWitt (1995) reported a Nexis search of newspapers, magazines, and television transcripts on the word "cyber" that turned up 167 mentions in January 1993, 464 in January 1994, and 1,205 in January 1995. Daily newspapers now run regular columns providing information and advice to prospective users of the Internet and other computer-based resources.

### 1.2.6 Information Access and Exchange

The use of computer networks for the dissemination of scientific information has been growing steadily. Research scientists, especially those located at universities, were among the first professionals who, as a group, used computer networks to distribute information on a large scale. Distribution techniques have been informal, for the most part, but today news of major developments in many fields typically is transmitted almost instantaneously to the major players in those fields via networks, and much of the scientific debate that once occurred at a leisurely pace on the pages of conventional journals now takes place at a much accelerated rate in cyberspace long before any of it finds its way into print media.

What this method of information distribution will eventually mean for the traditional peer-reviewed scientific journal remains to be seen. The idea of electronic journals that preserve some of the benefits, such as quality control, of the traditional system, but that greatly increase the speed with which findings become available to the research community and the ease of access to "publications," and that decrease the cost of that access, is receiving increasing attention and experimentation (Taubes, 1996). The 1994 edition of the *Directory of Electronic Journals, Newsletters* and *Academic Discussion Lists* listed 440 such facilities (Stix, 1994). According to a Science article, over 100 peer-reviewed science, technical, and medical journals were available on the Internet by the end of 1995. Increasingly, too, mainstream newspapers, periodicals and popular magazines are making their publications available electronically through CompuServe, Prodigy, WWW, or other information services in increasing numbers.

How are people to cope with the plethora of information riches that technology is beginning to provide, especially in view of the fact that all too often the treasures are completely commingled with an endless mass of infojunk and empty pointers? How will this technology change the role of the library, and of the librarian? What new information brokering services will it spawn? What sorts of new filtering and filing schemes will users need?

In February 1993, the U.S. Government announced a "High Performance Computing and Communications Technology Initiative" intended to be a major force in the development of a new National Information Infrastructure (NII). The vision for the NII is that it "will consist of computers and information appliances (including telephones and video displays), all containing computerized information, linked by high speed telecommunication lines capable of transmitting billions of bits of information in a second (an entire encyclopedia in a few seconds)" (OSTP, 1994, p. 8). The vision includes "[a] nation of users...trained to use this technology" (p. 8). "This infrastructure of 'information superhighways' will revolutionize the way we work, learn, shop, and live, and will provide Americans the information they need, when they need it, and where they need it—whether in the form of text, images, sound, or video. It promises to have an even greater impact than the interstate highways or the telephone system" (p. 8).

Targeted application areas for the HPCC and NII include several "National Challenge" problems, which are "major societal needs that computing and communications technology can help address in key areas such as the civil infrastructure, digital libraries, education and lifelong learning, energy management, the environment, health care, manufacturing processes and products, national security, and public access to government information" (p. 2).

This is a very optimistic vision. We believe that the degree to which it is realized will depend, in no small measure, on the care with which fundamental questions of HCI are addressed. We need to understand much better than we now do about how to ensure that the technology and tools that are developed are both useful—in the sense of meeting human needs—and useable. As a consideration of the question of how information technology has affected productivity to date will show (see below), it is not safe to assume that making powerful technology available guarantees that the anticipated desirable effects will follow.

### 1.2.7 Summary of Expected Trends

Many of the expected trends for the foreseeable future are continuations of those that have been apparent for some time. With respect to computer hardware, these include continued decreases in component size, power requirements, and unit costs accompanied by further increases in speed, reliability, bandwidth and storage capacity. Computing resources will be increasingly distributed, as will computer-based processes and the control of them. Applications and uses of this technology will continue to multiply and to become evermore diverse.

What is being called the "information age," or the "fifth wave," is expected to be characterized by a decreasing emphasis on heavy industry, mass production, and long-term stability of processes and products and an increasing dependence on flexible small-lot manufacturing, a greater emphasis on services, and growing recognition of information as a major commodity. Television, the telephone, and other communication media are expected to become increasingly integrated with computer network resources. More and more functions are likely to make use of, and become dependent on, information technology. The moving of bits in the form of electrons and photons could replace, to some significant degree, the transporting of people and material, although the extent to which this possibility will be realized will depend on factors other than technical feasibility. For example, will, or under what conditions will, people—basically social creatures—find substitutes for physical togetherness desirable?

There are many questions that are impossible to answer now regarding how information technology will impact the quality of life in the future. Some of these questions have to do with aspects of work and life that traditionally have been considered within the domain of human factors science and engineering—the designs of the functions and interfaces of the information systems that people will use, the effects of the introduction of new technologies in the workplace on work efficiency and worker satisfaction, the allocation of functions between people and machines. Many involve social, political, legal, or ethical matters that have psychological aspects and represent challenges both to system design and to applied cognitive and behavioral science broadly defined.

## 1.3 Human–Computer Interaction

### 1.3.1 A New Area of Research and Engineering

Human–computer interaction did not exist as a field of scientific inquiry or engineering fifty years ago—not surprising given the fact that electronic digital computers hardly existed fifty years ago. In the earliest days of computers, HCI was not a topic of interest, because very few people interacted with computers, and those who did generally were technical specialists. Excepting a few visionary essays, like that of Vannevar Bush (1945), papers on the subject began to appear only in the 1960s. Several seminal articles were written by J. C. R. Licklider (1960, 1965, 1968; Licklider and Clark, 1962) and Douglas Englebart (1962; English, Englebart and Berman, 1967).

Having gotten a start with papers like these, interest in HCI began to grow; as more and more people found themselves using computers for a broadening variety of tasks, the topic soon became an important focus of research. Articles appeared with increasing frequency in *Human Factors*, *Ergonomics*, the *International Journal of Man–Machine Studies*, *IEEE Transactions on Human Factors in Electronics* and elsewhere. The journal *Human–Computer Interaction* was founded in 1985. Other subsequently founded journals that focus primarily if not exclusively on this topic include *Interacting with Computers*, and the *International Journal of Human–Computer Studies*.

HCI has now been, for some years, a major area of research in computer science, human factors, engineering psychology, ergonomics, and closely related disciplines. Annual conferences are devoted exclusively to the topic, articles on it are regularly published in many broad spectrum journals—Baecker and Buxton (1987) estimated that by 1987 over 1,000 papers relevant to the field were being published annually—and it is the special interest of groups within traditional societies and a standard colloquium topic at professional meetings attended not only by computer scientists, human factors specialists and research psychologists, but by anthropologists, physical designers and other specialists as well.

### 1.3.2 Why Study HCI?

The argument could be made—the argument has been made—that research on HCI is not only unnecessary, but futile. The technology is moving so rapidly, according to this argument, that research has little chance of affecting it. By the time the results of research find their way into peer-reviewed journals, they are no longer of use to system designers. The systems for which that research was relevant are already obsolete. The only approach that will work in such a fast-moving area is a Darwinian one: throw lots of designs into the arena and let competition in the marketplace sort them out.

One need not accept or reject the argument—although we reject it—to note that it assumes that the only reason for doing research on HCI is to influence the design of future systems. We believe that that is one motivation for research in this area, but not the only one. First, we believe research can have its most significant effect on future design not by comparing designs or design elements, but by revealing the aspects of human tasks and activities most in need of augmentation and discovering effective ways to provide it. We believe that such research can be fundamental, and foundational to the synthesis and invention of computer applications that simply would not occur by intuition or trial and error.

Another reason for doing HCI research is to increase our understanding of the technology and its effects, to discover what impact computers (or uses of computers) are having on people's productivity, job satisfaction, communication with other people, and the general quality of their lives. Another is to increase our ability—although this will always be limited—to anticipate, and steer, future developments and effects. We think another important reason for studying HCI is to have some influence not just on the continuing development of the technology—to increase its usefulness and usability—but also, and perhaps more importantly, to help increase the chances that it will be put to uses that are constructive and humane.

What should be the goals of HCI research? We would like to know what makes systems more of less easy to learn to use, and what makes them more or less effective in the hands of experienced users; we would like to know how they could be more useful to people, more helpful in people's work, and more effective in facilitating their avocational and recreational pursuits. We would like to understand better what effects, good and bad, this technology could have on our lives and on future generations, and what can be done to maximize the desirable effects and minimize the undesirable ones.

It is generally believed that this technology, already powerful, has the potential to change life on the planet profoundly. We need to know how to think about it and its potential effects. We need to know how to relate it, in our reflections and planning, to other major agents of change. We need to understand better its potential implications for world peace, for environmental change, for fighting the age-old problems of disease, poverty, and political repression. We should be concerned too to understand its potential for criminal, antisocial and inhumane applications. Only by understanding the various possibilities can we hope to influence effectively which of them are realized or to what extent.

### 1.3.3 Productivity

Among the many reasons for studying HCI, a particularly compelling one relates to productivity—a primary concern of any industrialized society, because it is believed to be a major determinant of a country's ability to compete in world markets and of the general standard of living that its citizens can maintain. Technological innovation is considered by many economists to be the most effective means for increasing productivity (Baily and Chakrabarti, 1988; Cyert and Mowery, 1989; Klein, 1988; Mokyr, 1990; Solow, 1959; Young, 1988). Many corporations have made huge investments in information technology on the assumption that such investments would more than pay for themselves by the productivity gains that would result from them. Unfortunately, there is compelling evidence that computer applications have not had the expected general and major beneficial effects on labor productivity.

#### Bad News

It appears that, despite the enormous gains in hardware power and software versatility, and the widely held belief that computers are very effective productivity tools for business and related applications, the efficiency with which everyday humans perform everyday tasks has not been greatly improved by computer aids. On the contrary, the evidence is that, on average, the effects of computer aids on worker efficiency and economic productivity is relatively slight. A detailed review of a wide spectrum of data that converge on this conclusion has been provided by Landauer (1995, pp. 1–78). We will summarize only a few highlights here.

First, national economic productivity, measured as constant dollars' worth of output per hour of labor expended, has grown less than half as fast since the introduction of widespread computing than in any previous period for which data exist. Dating the widespread use of computers from the mid–1970s, by which time, for example, over 90 percent of banks were using them heavily, by 1995 over $5 trillion had been spent in the United States on computing hardware, software, and support. Meanwhile U. S. productivity grew at a little over 1 percent a year, compared to 2–3 percent annually over the previous century. The same story repeats itself in slightly altered form in all the other major industrialized nations. The picture is darkest in the service industries—banking, insurance, restaurants, transportation...—where most information workers are found. Although it is often argued that the problem is with the measurement of productivity in services, most of the "service" industries involved actually are subject to reasonable measurement, and economists who have studied the situation do not believe that much of the deficit in productivity growth can be accounted for this way. Moreover, productivity in service industries had been increasing steadily by the usual measures before their move into computers as their main capital investment (for a thorough review of this issue, see Landauer, 1995, pp 95–101).

Other data regarding computers and productivity include comparisons among major industrial groups, like banking, insurance, communications, transportation, and retail stores. Those groups that have invested more heavily, that have provided their workers with the greater amounts of information technology, have not fared better, generally speaking, in terms of either productivity as usually measured or business success indexed by return on investment, than have those that did less in this regard. The same holds for particular firms within major industries: firms that computerized vigorously during the 1980s and early 90s showed no better returns, on average, than those that did not. Indeed, with some exceptions, econometric analyses, of which there have been about a dozen, show either slightly negative effects of computerization or very small positive gains. This situation can be compared with similar analyses for investments in buildings and machinery, which, on average, show returns of about 13 percent. Several economists and management experts have commented that most businesses would apparently have been much better off had they been able to invest their capital in bonds at market prices instead of computers (e.g. Baily and Chakrabarti, 1988; Franke, 1987; Loveman, 1990; Strassman, 1990).

But caveats to all this must be stated. Manufacturing was becoming a smaller and smaller part of the advanced economies and service industries a larger part during the period mentioned. Service industries have always had lower productivity than manufacturing, as traditionally measured, and slightly lower productivity growth. Thus, there was no necessary reason why productivity should have continued to grow at historical rates as the mix of industries changed. It could be that computerization has allowed movement into services as manufacturing declined, forestalling an even greater potential stagnation of productivity growth. In short, the fact that productivity has increased at a slower rate since computers became prevalent in the workplace than it did before does not prove that the introduction of computers was the cause of the change; on the other hand, the fact that the increased rate of productivity growth that many industrialists anticipated was not realized challenges the assumption that large expenditures for computing resources are easily justifiable in terms of resulting productivity gains. Of course, there may be other justifications, and there are important psychological and cultural value questions to ask about whether economic productivity is the right or most important goal of progress, matters on which we will comment later.

Whether computers have caused a slowing of productivity growth, or have mitigated what would otherwise have been a worse situation, they have brought no great leap forward in productivity and in standards of living comparable to those caused by previous technological revolutions (which often also accompanied major shifts in economic activity—think of what happened in agriculture) Revolutions that exploited new sources of inanimate power, new mechanical and chemical technologies, new agricultural techniques, and, more recently, technologies for communication and transportation, have had major overall economic consequences. For example, a significant improvement in national productivity can be traced quantitatively to introduction of the electric motor alone (David, 1990). Information technology and computers certainly seem to be just the right technologies to multiply the efficiency of service work, yet it appears that they have not done so.

The most discouraging and, to HCI professionals, challenging, evidence has yet to be mentioned. This is the fact that relatively small gains in individual work efficiency have resulted when people have moved from old ways to new computer-aided ways of doing things. Word processing, for example, is the most widespread use of computers in service industries, with the possible exceptions of point-of-sales transactions and inventory control. Text production work has been studied both in the laboratory and in the field. Generously credited, there is at most a 50 percent increase in the amount of work produced per hour (Landauer, 1995 pp. 50–56). While this may seem a lot, such an efficiency advantage is often not enough to pay for the increased equipment and the organizational and technical support, and educational efforts, required. Moreover, whether it can be considered increased efficiency, in strictly economic terms, is questionable when one takes into account the fact that people who operate word processors are paid at a higher rate than those who simply type.

Many more drafts of a document are likely to be produced with word processing resources than without them, and much more paper is likely to be printed, but evidence of greater quality or markedly quicker end-to-end throughput is missing. There have been several studies of the quality and time taken to compose business letters and student essays with and without word processors (e.g., Card, Robert and Keenan, 1984; Gould, 1981; Vacc, 1991). The box score slightly favors paper and pencil for quality, and shows mixed results for time efficiency depending on user skill and the type of application. Often word-processor technology is used to move document preparation from document specialists—typists, compositors, etc.—to highly paid professionals with lower document preparation skills who would otherwise be more productively employed.

The work aids brought in by other technological and economic revolutions had hugely greater effects. The equipment for spinning thread produced a 30,000 percent (i.e. a 300 to 1 ratio) output-per-labor-hour improvement over a period of approximately the same length of time that computers have been around (Mokyr, 1990). Even since computers have been here, the productivity of textile manufacture has gone up by over 200 percent. Total gains in output per worker hour in agriculture were on the order of 3,000 percent before the computer era and have continued to grow at about the same rate since.

Thus, the size of the work efficiency gains attributable to information technology and computing are tiny compared to those that have been the engine of progress in modern industrialized societies. It takes much more than a 50 percent or even 200 percent gain in individual efficiency to translate into great improvements in overall productivity of a single company, or an industry, or a nation, because individual improvements, which generally apply only to parts of a business, have to be effectively deployed and woven into organizational fabrics (National Research Council, 1994). And gains happen only when technology is actually applied to increase productivity; much computer power, for example, has gone into purely competitive activities, like security market manipulations, complex insurance and investment instruments, and airline seat marketing strategies, that shift business share from one firm to another but do not materially increase total output value.

### Good News and Challenge

The challenge to HCI is to help change the bad news to good. Usefulness and usability are where many basic problems lie, and this is the province of HCI. Computers are spectacularly wonderful machines, extraordinarily well suited to being the basis of better tools for intellectual—thus service—work, and for a host of yet-to-be imagined ways to enhance everyday life. We need to find out why they have not been having a greater beneficial impact and how to make them do so.

Landauer (1995) suggests that the root cause of the problem is an inadequate engineering design discipline for applications software. In particular, the usefulness and usability of software have usually not been subjected to the regular and systematic testing of actual utility that underlies improvement in most other technologies. Evaluation has concentrated on the internals of the systems themselves: how fast they calculate, how much data they store, how flawlessly they run, how many features they support, the quality of their graphics, the impressiveness of their tricks. Too often, insufficient attention has been paid to measuring how well and in what ways systems do and do not actually help people get work done. Even when usability evaluation is extensive, it rarely measures or aims directly at improving work efficiency, it is primarily concerned with superficial aspects of ease of use as they will affect sales appeal. Commercial development efforts have almost never compared the efficiency of people using computer methods with those using state-of-the art noncomputer techniques.

The good news is that techniques exist for improving things and using them can and does make a great difference. Gathering together all reports of development projects in which software intended to aid a job has been informatively evaluated, redesigned and tested again, disclosed stunning results. The average gain in the efficiency of work done with the help of a computer system as a result of one short cycle of empirical user-centered design is over 50 percent. Empirical user-centered design means tests with real users doing real work, or some equivalent assessment method that allows observation of what the strength and weaknesses of the technology are for its real users' real purposes (see, e.g. Landauer, 1995 p 221–227; Nielsen 1993 and many chapters in this volume.) From data of experiments by Nielsen (Nielsen and Landauer, 1993), it can further be shown that the benefit-to-cost ratio—where the benefit is for the end users—from such activities is simply enormous, almost always over 20-to-1, and often in the 200-to-1 range. Moreover, to get substantial effects often requires very minor amounts of effort, only days of effort by a few people (Nielsen, 1989). Thus, to put it bluntly, there is a bone-simple way to improve products immensely.

The immediate challenge, then, is to get user-centered design—empirical evaluation and observation and iterative improvement—actually in place so that it can do its job (Nickerson and Pew, 1990). The result could be much more than improvement of individual products. It could also bring us closer to the general principles and methods that could eventually help us to design better in the first place. Bridge designers can do good initial designs from first principles. However, they have learned the engineering science behind their craft largely from analyses of successes and failures over a long period of time. Bridge designers have the advantage of a relatively obvious criterion of success and failure. For mind tools, effective feedback requires special assessment methods and effort. But regular practice of iterative formative evaluation would provide an enormous amount of the needed data. The challenge facing HCI is to be able to use knowledge about actual effectiveness in helping people achieve their goals to make new applications not only novel, exciting, and technically impressive, but useful, usable and socially valuable as well.

We have stressed productivity here, because it is an issue of great concern, and rightly so, we think. In its broadest connotation, increasing productivity means, among other things, getting more out of less—making more efficient use of resources. This seems like something we, as individuals and corporately, should want to do. On the other hand, we think also that this emphasis on productivity needs to be qualified in two ways. How productivity is best measured in a "post-industrial" economy is a persisting and difficult question. Second, we do not mean to argue that increasing productivity should be our only, or even necessarily our major concern. It would be a dubious gain if we managed to get everyone producing more and enjoying life less. Presumably increased productivity is considered a desirable goal because it is assumed to be a way to raise the standard of living—to enhance the quality of life—not only for those people in the segment of an economy whose productivity is at issue but for others as well. It seems a reasonable assumption, but we need to consider the possibility that there are conditions under which it would not be true.

This is a complicated subject and we do not propose to pursue it here, beyond noting that a major determinant of the quality of workers' lives is the satisfaction (or dissatisfaction) they get from their jobs. When productivity is increased at the expense of making jobs less interesting or intrinsically less rewarding, it is not clear that progress has been made. We think that the enhancement of job quality should be a major objective of HCI along with increasing productivity.

More generally, as mentioned earlier, we need to go beyond economic measures of the effects of technological advances to assess their effects on the quality of workers' and consumer's lives and their influences on families, communities and nations. Some research effort has been exerted in this direction, as well as considerable popular speculation. The research literature has tended to be inconclusive (see Attewell and Rule, 1984, for a survey, Kraut, Dumais and Koch, 1989, for an exemplary controlled study), and the opinion expressers fiercely divided (for example compare Clifford Stoll's, Silicon Snake Oil, and the Tofflers' Third Wave). Our own conclusion is that much more thought and research on these matters is needed. HCI researchers and designers have a major role in choosing what results to strive for and to assess. We would not maintain that the economic efficiency gains that have motivated much of the computer revolution are either hopeless or unworthy, but we would urge that other goals—the augmentation of human competence and satisfaction, and of societal harmony—are of at least equal importance.

### 1.3.4 Increasing Importance of Topic

We believe that, as a research and design activity, HCI will become increasingly important. This belief rests on some of the expectations mentioned above, including a continued decrease in the cost of computing hardware, continued expansion of computer networks and increasing access to them, increasing use of computer technology in the workplace, increasing availability of software that is potentially useful or entertaining for the general consumer, increasing use of computers in education, and a growing assortment of on-line services available through computer networks. The true value of all of this will depend critically on the quality of the interaction of humans with computers.

In short, the size of the population of computer users and the variety of uses to which information technology is put are both likely to grow rapidly for the foreseeable future, as they have been doing in the recent past. The challenges to research and design are many and extraordinarily diverse. Although the topic has already received considerable attention, we believe that the surface of possibilities has hardly been scratched, that many of what will prove to be the most significant challenges have yet to be conceived, and that the field could benefit not only from the articulation of some new questions but from the development of some new approaches to investigation and innovation.

## 1.4 How to Study HCI?

How can research help to push HCI forward? The literature review cited earlier demonstrated that systems that have the benefit of evaluation and redesign are usually much better than those that do not. It seems clear, then, that some form of research can guide progress. What kinds of research are practicable and effective in this field? The general endeavor is what is usually called applied research, yet there does not appear to be a great deal of relevant basic science ready to be directly applied. Perhaps it is more appropriate to HCI (as to many other fields of invention and design) to speak of "engineering research," meaning investigations into what functions to build for desired effects and how to build them. Among published accounts of successful efforts to improve the objectively measured utility of computer applications, many research methods are described; indeed, unless one classifies the methods quite broadly, no two projects appear to have used exactly the same method. Adopting one such broad categorization, we give the following partial list of techniques that have been involved in significant achievements and that seem to us especially sound and promising.

### 1.4.1 Task Analysis

Task analysis is a loose collection of formal and informal techniques for finding out what the goals are to which a system is going to be applied, and what role new technology might play in helping to achieve them. The center of attention is not the computer system, but the goals of the activity (business, education, entertainment, social interaction...) that is contemplated. Indeed, the best attitude is to assume that non–computer solutions are at least equally welcome, and to let analysis and imagination roam freely over possible, and maybe even seemingly impossible, approaches.

The first step in task analysis is always to go to the place of work and see what is being done now. Analysts watch people and ask questions, talk some to executives, managers, and marketers, a lot to the people doing the activity. The way work (or play) is actually done is seldom the way it is officially prescribed. Many analysts try to get people to talk aloud as they go about what they are doing. The context brings things to mind that are not thought of or mentioned in an interview. In a technique sometimes called video ethnography, analysts make inconspicuous video tapes that can reveal how work is actually done and how it depends on social norms and processes. Footage of struggling workers can be especially valuable for converting unbelieving executives and designers. Analysts measure how long people spend doing specific tasks and notice what kinds of errors they make. How much time does each subprocess take? What contributes to variability in performance? Are some workers faster than others, some kinds of tasks finished more quickly than others? When speed is a priority, how might slow jobs or slow people be converted into fast jobs and fast people?

### 1.4.2 Consultation of Potential Users

When it comes to jobs, it is hard to be wiser than the people doing them, perhaps not wise to try. What is sometimes called the "Scandinavian School" of system design prescribes going out into the work environment in which a system is to be used and bringing users into the process at every step. The emphasis is on personal, social and organizational factors in the introduction and use of new technology. While one might not ordinarily think of such activities as research, in the HCI context they are. Traditionally, computer applications have been largely, sometimes entirely, technology driven. Seeking knowledge about desirable characteristics of systems from a source other than the technically sophisticated but application-naive developer's perspective is thus an important kind of research. Often the naturalistic data that can be acquired from workers simply by asking is sufficiently precise and reliable for the purpose at hand. Improving understanding of the organizational, social and motivational milieu in and for which an information system is to work is one such purpose.

A caveat is necessary with respect to both task analysis and user consultation. The introduction of information technology in the workplace often changes the nature of the jobs that are performed in unanticipated ways. Although the intent may be to increase the effectiveness with which one can perform a particular task, the result may be to modify, perhaps profoundly, the tasks that the worker will perform and the interpersonal, social and business processes of which it is part. Analysis of an existing task may fail to provide a clue to the transformation that the new technology will effect, and neither workers nor managers may be able to imagine the ways in which activities and relationships will change before they have had any experience using that technology. What this suggests is that task analysis and user consultation should be thought of, not as activities that are carried out once at the beginning of a design project, but as dynamic processes that must be ongoing, in one or another guise, over the entire course of the design and development effort and over the life of the system's deployment. This is part of what the notion of "iterative design" involves.

### 1.4.3 Formative Design Evaluation

The term "formative evaluation" originated in the development of instructional methods, where it is contrasted with "summative evaluation." Formative evaluation is used to guide changes, summative to determine how good something is. Unfortunately, in the past most usability testing for computer systems has been merely summative, and rarely produced useful design guidance or generally applicable principles. The idea of formative evaluation is not to just decide which is better, A or B, but to learn what does and does not work and why.

The need for formative, as opposed to summative, evaluation in computer system development has been recognized by many writers (Carroll and Rosson, 1984; Eason, 1982; Gould, Boies and Lewis, 1991; Sheil, 1983), and it stems from the fact that computer systems and user interfaces are compounded of myriad interacting details any one of which can potentially defeat or enhance usability and utility, while few of their effects can be reliably anticipated, and also, in part, from the fact, just mentioned, that new technology can change tasks in unanticipated ways. The idea that both exploration and evaluation need to occur more or less in parallel throughout the development process is reflected in the notion of "guided evolution" (Nickerson, 1986).

### 1.4.4 The Gold Standard: User Testing

Only if we study real workers doing real jobs in real environments, can we be sure that what we learn is truly relevant to design. User testing tries to get as close to this ideal as is practical by testing people like those for whom the system is intended with tasks like those they will most often do, using a mock-up, a "Wizard-of-Oz" simulation in which a human plays the role of system, or a prototype or early version, either in situ or in a laboratory setting that is similar to the intended office, home or other place of use. Experience suggests that such compromises are usually, if not always, good enough. Information gained from user tests has been the most frequent source of major usability improvements. User testing is straightforward. Users try, the tester watches, notes errors, times tasks, later asks questions. Many practitioners urge the users to talk aloud as they work. Some make videotapes to review and analyze in detail, although there is debate about the cost-effectiveness of this time-consuming process. On initial design, the average interface has 40 unsuspected but identifiable flaws. Testing two trial users on average reveals half of the flaws, tests with four users reveal three-quarters, and so forth (Nielsen and Landauer, 1993). Most of the flaws are fixable, and observing the test users often leads to creative insights that feed innovation.

### 1.4.5 Performance Analysis

The goal of performance analysis is deeper understanding on which to base innovation and initial design rather than assessment and improvement of existing designs. Broadly speaking, performance analysis studies people doing information-processing tasks in an attempt to understand what they can do well and what they typically do poorly, where help is needed and, if possible, how that help might best be provided. Performance analysis is done either in the laboratory with somewhat abstracted tasks, such as suggesting titles or key words for information objects, or with an existing technology for the performance of some real job. Performance analysis, when successful, leads to the identification of a human performance limitation that a computer can ameliorate. Computers are sufficiently powerful and flexible that finding the right problem is often harder than finding its solution. The unlimited aliasing technique described by Furnas, Landauer, Gomez, and Dumais (1987) illustrates the point. Once it had been discovered that people typically refer to "information objects" such as files, records and commands, by many more names than systems usually recognize, providing an effective solution—many more "aliases"— was straightforward.

Performance analysis pays special attention to time, errors, and individual differences, all of which can provide important clues to deficiencies in present technologies and processes and opportunities for improvement. In computer-based tasks, mistakes often lead to extreme difficulties that are very costly in user time, lost work, propagated erroneous information or irretrievable records. Computers also often greatly amplify the differences in efficiency between different people (Egan, 1988.) Studying the sources of errors and the abilities that are demanded can lead to insights on which HCI progress can be based.

### 1.4.6 Science

It is tempting to think that fundamental research in psychology might provide the basis for improving HCI. To some extent it can and already has, and one hopes that eventually it will do more. But we believe the extent of direct help is modest at the present time. Psychology has provided some real advances in understanding human behavior, but only in limited domains, usually represented by relatively narrow problems that have been brought to the laboratory. A few well established fundamental principles speak to system design: the Hick–Hyman law, according to which decision time is proportional to the log of the number of equal alternatives, implies that a single screen with many choices is better than a series of screens with a few choices each (Landauer and Nachbar, 1985); Fitts' law, which tells how long it takes to point to an object depending on how large and far away it is, helps in designing pointing devices and laying out mouse targets on a screen; the power law of practice, which tells how response speed increases over time, can sometimes predict how well experts with a new system will perform relative to those using an old system; the facts of working memory limitations suggest that users should have to keep very few new, unorganized, pieces of information in mind at once; human factors principles and tables for color combinations and font size/distance requirements help designers avoid suboptimal displays; and there are more examples. A noteworthy effort to apply the results of basic research in the HCI domain is the work of Card, Moran and Newell (1983).

A fairly representative example of both the success and limitations of basic science in HCI can be found in early efforts to provide underpinnings for the design of command languages. Are "natural" words and syntax an advantage? Experiments found that for small systems, say a basic text editor, the choice of words , e.g. using "allege," "cipher" and "deliberate" instead of "omit," "add" and "change" was inconsequential ( Landauer, Egan, Remde, Lesk, Lochbaum, and Ketchum, 1993; Landauer, Galotti and Hartwell, 1983.) Using "normal" English word order sometimes made some difference, but not a great deal. Using a consistent word order—verb before object, for example—sometimes was helpful, but not universally so (Barnard and Grudin, 1988; Barnard, Hammond, Morton, Long, and Clark, 1981.) The user–language research with the best payoff was on how to construct abbreviations; it turned out that using a consistent rule was more important than what rule was used (Streeter, Acroff, and Taylor, 1983).

These and other examples show that research aimed at better understanding of the cognitive underpinnings of the use of information systems can pay off. However, it will take time, and much greater volume than the current trickle, to make a major contribution. We believe there is much to be gained by applied research aimed more specifically at developing a better understanding of the complex dynamics of interactions between human intellects and powerful information-processing computers, inasmuch as it is just these highly complicated and nonlinear characteristics of HCI that it is most important to accommodate in system design. In the short term, we are more optimistic about the opportunities for direct engineering research than about the promise of new basic research creating a significantly useful applied science of HCI. On the other hand, we recognize that research that is focused narrowly on the details of a specific system's design may lack generalizability. There is a happy medium: research that is addressed to specific aspects of HCI but that is sufficiently generally conceived to have applicability across many systems, and that is done within the conceptual framework of some psychological theory so that it can have theoretical import as well.

Before leaving this topic, it is important to note that the domain of HCI has sometimes provided a fertile ground for the elaboration and testing of cognitive theories, a development in the opposite direction from "applied science" as usually conceived, but a direction that is not unusual in science and technology (Kuhn, 1977; Mokyr, 1990). Good examples include the work of Card, Moran and Newell cited earlier, that of Polson and Keiras (1985) on production system models of problem solving and transfer, and of Singley and Anderson (1989) and Pennington and her colleagues (Pennington, Nicholich and Rahm, 1995) on transfer. Another recent example is the progression from the studies and innovations in information retrieval methods at Bellcore to a new model of human word learning and memory representation (Landauer and Dumais, 1996).

## 1.5 Some Specific, and Some Broader Issues

The psychological and social aspects of HCI are many and diverse. Questions of interest run the gamut from those pertaining to the layout of keyboards and the design of type fonts, to the nature of effective navigation aids and information displays, to the configuration of virtual workspaces that are to be shared by geographically dispersed members of a work team. They involve the effects that computer systems can have on their users, on work, on business processes, on furniture and building design, on interpersonal communication, on society and social processes, and on the quality of life in general. Some research is motivated by an interest in making computer-based systems easier to use and in increasing their effectiveness as tools; some is driven by a desire to understand the role, or roles, this technology is likely to play in shaping our lives. Here we can do no more than comment briefly on some of the specific issues we think will be among those most important to consider for workers in human–computer interaction in the near-term future.

### 1.5.1 Equity

The question of equity in relation to computer technology takes many forms. Much of the discussion of the question has focused on economic aspects of the issue. Will computer technology be differentially available to the haves and the have-nots, where this distinction sometimes pertains to individuals and sometimes to nations? Will the increasing globalization of network facilities give third-world scientists greater access to the rest of the scientific world, or will it increase the isolation of those in countries lacking the communications infrastructure on which ready access to the Internet depends (Gibbs, 1995)?

Questions of these sorts have been of concern for some time, but they have not been a burning issue. As long as the large majority of people did not have computers and ownership of them was not considered to be especially beneficial, the nature of the concern was the possibility that a small minority of people had some advantage over the majority and the likelihood that that small minority did not include a proportionate share of low income people or other disadvantaged groups. The concern is likely to increase considerably as more and more of the people who can afford to purchase computers do so and the use of these machines plays increasingly significant roles in their daily lives. If access to computers and computer-based resources becomes a critical determinant of how effectively one can function in society, as it seems almost certain to do, the question of equity will become a very serious one indeed. In the U.S. the disparity in computer ownership and use between the richest and poorest quarters of the population was almost eight to one in 1993, and had grown in the previous five years. Differences between urban and rural dwellers, those over 60 or below, and those with or without high-school educations, and of differing ethnic or racial origins were less pronounced, although still large, and had also grown (Anderson, Bikson, Law, and Mitchell, 1995).

There is another sense in which equity is an issue, that has not received much attention, and this is the question of the possibility that computer technology will differentially benefit the can-dos and cannots. "Cannots" in this context refers to individuals who, perhaps because of lack of training or limitations of ability, are not able to make effective use of a computer system even if one is available. Will this technology tend to be an equalizer or to amplify differences in knowledge and ability? Maybe it will create new opportunities for everybody, in principle, but will also, in practice, give an enormous advantage to those who are capable of making the most of those opportunities. If this is what it does, is that bad? How one answers this question is likely to depend on one's system of values and one's concept of justice. But we suspect that most people would agree that one worthy goal would be to find ways to ensure the usefulness of computer technology to people who have to cope with one or another type of disadvantage or disability.

### 1.5.2 Security

Schiller (1994) points out that much of what eventually became the Internet was designed with only trustworthy users in mind and that, consequently, privacy-invading intrusions on network communications and resources are not difficult. As the numbers of individuals and institutions using network facilities to work with and transmit sensitive information have steadily increased, efforts to develop measures to ensure security have increased apace. Encryption techniques are now commonly used for data transmission, and safeguards have been implemented to restrict access to private files. None of the approaches that have been developed is foolproof in all respects, however, and it seems unlikely that any of those that will be implemented soon will be. The possibility of unauthorized access to information residing in a computer that is connected to a network will remain a fact of life for the foreseeable future.

Much of the current debate about security and what should be done about it focuses on the question of what constitutes an acceptable compromise between security and freedom of access (Germain, 1995). Ideally, one wants to maximize freedom of access by legitimate users of systems and databases while making access by intruders impossible, but the two goals may not be simultaneously realizable. To prevent access by intruders, or even to impede such access significantly, it may be necessary to inconvenience legitimate users to some extent as well; assuming this to be the case, the question is how to make this tradeoff in an acceptable way.

Impersonation and forgery are special problems for electronic mail. As Wallich (1994) puts it, "virtually no one receiving a message over the net can be sure it came from the ostensible sender" (p. 91). "A message from 'president@whitehouse.gov' could as easily originate from a workstation in Amsterdam as from the Executive Office Building in Washington, DC" (p. 99). Technologists are keen on finding a solution to this problem that does not impede freedom of access to the net.

### 1.5.3 Function Allocation

There are many issues relating to the question of who (or what) is to do what in a system that includes both humans and computers. At one time it was fashionable to construct lists showing which functions could be performed better by people and which could be done better by computers. This is a somewhat risky venture, however, because the capabilities of computers have been changing fairly rapidly, whereas those of human being are little, if any, different from what they were 50 years ago.

As it happens, many functions can be performed either by people or by computers; more specifically, the list of things that people can do and computers cannot has been growing smaller over time and is likely to continue doing so. This does not mean that people are going to have less and less to do; that would be the case only on the assumption that if a computer can be made to perform some function, it *ought* to be given that function to perform.

We believe that the question of what tasks should be assigned to computers is of fundamental importance and that it will be the subject of debate more and more as the capabilities of computers continue to expand. The basis on which the decision should be made will itself be a matter on which opinions are likely to diverge; what is clear is that whether or not a function is within a computer's capability is only one of the factors that will figure in the debate. What functions do we want computers to perform? What do we want them to do with our help or guidance? What do we want them to help us do? And what do we want to do without their involvement? What do we want to be made easy for us, and how easy, and what do we want to remain a challenge to special skills that grant satisfaction in their execution in part because of their difficulty; we think of such skills as musical performance, art and chess? Such questions can only grow in importance as the technology and the potential it represents continue to advance.

### 1.5.4 Effects and Impact

We have already noted that companies have spent a great deal of money to acquire large amounts of computing power in the expectation of realizing cost-cutting efficiencies and increases in productivity, and that evidence that the expected benefits have been realized is at best equivocal. One hypothesized reason for the failure of corporate investments in computing resources to have yielded the expected returns is mismanagement, and, in particular, the tendency by managers to acquire much more computer power than their companies know how to use effectively (Leutwyler, 1994). However, we believe that even more important is lack of usefulness and usability, caused by failures to evaluate and design systems well enough to produce their intended effects of enhancing the ease, efficiency and waste-free coordination with which work is accomplished by individuals and organizations (Landauer, 1995).

The issue of productivity aside, there can be little doubt that the introduction of computer technology in the workplace has changed the nature of many jobs, including many that have—or had—little if anything to do with technology. At least as important as the question of the effect of these changes on productivity is that of the effect the new technology has on people's satisfaction with their work. Given that work is not only a means of making a living, but an important part of living, it should be intrinsically satisfying, and we need to know much more about whether and how technologically driven changes in work tend to make it more so or less.

An area of application of information technology that many have long believed to have great potential, but that has not had the expected impact to date, is education. Many believe the potential is still there and even increasing as the technology continues to advance (Nickerson and Zodhiates, 1988; Tinker and Kapisovsky, 1992), and examples of what can be accomplished have been reported (Bruce and Rubin, 1993; Lajoie and Derry, 1993; Larkin and Chabay, 1992; Perkins, Schwartz, West, and Wiske, 1995; Psotka, Massey, and Mutter, 1988; Ruopp, Gal, Drayton, and Pfister, 1993). The success of efforts to apply information technology to education is likely to depend on a variety of factors, not all technological, but the extent to which systems intended for use by students and teachers are well designed from an HCI perspective must be among the more important of them.

As we have already noted, information technology is a resource of enormous potential for constructive and destructive uses alike. The same technology that can bring us unprecedented access to information, educational resources of many types, greater participation in debate about issues of local, national, or international interest and policy, aids to creative pursuits, personal contact with people around the world...has also the potential to bring us electronic surveillance, privacy invasion on a grand scale, crime and mayhem of epic proportions, degrading and exploitive forms of entertainment.... But all technologies can be used for bad purposes as well as good, and the more powerful the technology, the greater its potential for good and bad alike. It is a continuing challenge to people of good will to ensure that the good uses of information technology exceed the bad by a large margin.

Perhaps even more challenging than the problem of limiting the technology's "affordances" to people who intend to use it to work mischief is that of trying to anticipate and avoid undesirable effects that no one intends. This requires giving some thought to what the effects of various developments could be, and perhaps challenging assumptions that are sometimes made uncritically to justify moving the technology in one or another direction or using it in one or another way.

What, for example, will be the effect of efforts to engage the entire citizenry of a country in decision making on national policy issues through instant polling and electronically conducted referenda? Will government be more immediately responsive to the will of the people if democracy is made more participatory, or is it the case, as Ted Koppel (1994) has argued, that "nothing would have a more paralyzing impact on representational government?" Will interactive advertising help people make better informed purchasing decisions, or will it increase the ease with which advertisers can convince people to buy what they neither need nor really want? Might information technology make people become more isolated? Less likely to come into face-to-face contact? How fully can remote contact satisfy social and psychological needs?

Such questions, of which a very large set could be generated, are difficult, perhaps impossible, to answer in the abstract. It is possible, however, to collect data on effects as systems begin to be used for various purposes and thereby to gain some fact-based understanding of what the effects of specific uses are. This understanding should help guide further developments.

### 1.5.5 Users' Conceptions of the Systems They Use

Users develop mental models of the systems they use. These models may be consistent with reality to greater or lesser degrees. A common reaction of people interacting with systems that display some modicum of cleverness is to impute to these systems more capability or intelligence than they have. Observations of users of Weizenbaum's ELIZA (1966), a rather crude simulation of an interacting human by today's standards, revealed this tendency in sometimes unsettling ways. The complexity and richness of the world of information resources that is approachable from computer terminals today will support the development of very fanciful models of what lies at the other end of the wire. Models that fail to correspond very closely to reality may or may not cause difficulties of various types. It would be good to understand better the factors that determine what types of models users develop and how to facilitate the development of adequately accurate and useful ones. How one conceptualizes the world of information to which the technology provides access may determine how effectively one is able to navigate in that world; conversely, the types of navigation aids that are provided for users will undoubtedly influence the mental models they develop.

### 1.5.6 Usefulness and Usability

Usefulness and usability are the twin goals of HCI research and development. First and foremost a system must do something that is helpful or pleasing. Although it may bring sales success, in the long-run it is not enough that a system does something that is novel, clever or impressive; what it does should provide significant advantage over what we had before. The challenge of design for usefulness starts with understanding what people need, what they would like to do that they find difficult, and proceeds to finding ways to make the impossible possible, the difficult easy, the slow fast. How to take an individual or family, at their convenience, from home to workplace, relative's house or distant vacation land was the usefulness challenge of transportation. It was answered by the invention of trains, autos and airplanes. What do people want to do or do better, and how can computers help them satisfy this desire; these are the pressing questions.

Usability is the next issue. Once we have identified a useful function and learned how to provide it, we must find a way to make its operation easy to invoke and control and its output easy to understand, verify and alter; in short to make the computer's interaction with its human master serve the master rather than the slave. By now there has been enough experience in the design of interfaces and interactive processes to ensure the availability of scores of effective devices and techniques and of considerable expert judgment and creativity. Nevertheless, the complexity of humans and computers and their joint behaviors, depending as they do on natural cognitive processes in the one and artificial ones in the other, insures that first-try drawing-board solutions will rarely be good enough. Many approaches to the complexity of HCI design have been taken, and many are described in this handbook. They range from theory and modeling of thought and action sequences to methods for simulating human–computer transaction protocols and their outcomes. Progress and successes have come from such efforts, and more are to be expected. Still, at the end of the day, system designs, like bridge designs, must be tested and revised both early and often to assure usability. Among the techniques to achieve usability the most useful and promising are methods to simplify, speed and increase the accuracy of the iterative test-and-fix cycle, and ways to make test findings more helpful for future design.

It is clear that usefulness and usability are closely related, that it is hard to have one without the other. To be sure, extremely useful machines—we think of jet liners and backhoes—may justify long years of training of skilled operators, and there is a necessary tradeoff between utility and complexity. But as a general rule, making a tool easier to use will, *ipso facto*, make it more useful. And, given the rapidity with which computer applications multiply and change, thereby limiting the time available for learning, trying hard to maximize usability is not a luxury but a necessity.

Still, usefulness and usability, essential as they are, and even when synergistically combined, are not quite the whole story. Systems must be desirable as well. A technology will be of little benefit if few use it. Word processors were at first used to centralize document preparation. As a result secretaries lost their knowledge of office context and jargon, and could not fix problems by conversations with the authors of the documents on which they were working. Consequently, word-processing centers lost their popularity (Johnson and Rice, 1987).

A desirability failure that degrades the user's life is especially serious. Centralized word processing in businesses often has this unwelcome effect as well. Typists and their supervisors usually intensely dislike this style of work, its social isolation from the life of the office, its artificial work-flow control procedures, its separation from the real consumer. High on the list of issues to be addressed by HCI efforts should be that of making system design intentionally and systematically improve the lives of individuals and the functioning of the groups to which they belong.

## 1.5.7 Interface Design

### Displays

One aspect of hardware the future progress of which is particularly important to HCI is displays. Computer screens in popular use offer a much smaller viewing and interaction space than does paper and print technology. Although computers can partially compensate for this disadvantage by serial presentation of text and graphics in multiple overlapping windows, and by provision of dynamic displays not possible in print, they do not exploit very well the human's wide field of vision and action in space.

Thus, many desks that hold computers are still littered with paper, books and notes, and users have difficulty maintaining perceptual, memory and attentive continuity between components of complex tasks on the screen, e.g., the relation between entries in multiple spreadsheet pages or reference, text, footnotes and figures for a document. A back-of-envelope calculation suggests that a viewing space roughly four to six times that of current one-page screens would be desirable, if not optimal, because it would accommodate most of the space in which text can be read conveniently by moving just the head and eyes.

Display resolution is also still suboptimal. At normal reading distance, the human visual system can usefully resolve detail of about an order-of-magnitude greater density than current displays provide. More research is needed into the parameters of size and resolution for visual displays that are required to optimize computer-aided human performance.

Visual displays have been the predominant means of communicating from computer to user in the past and are likely to continue to be so for the foreseeable future, if not indefinitely. This should not obscure the fact, however, that the possibilities for communicating via other modalities—especially hearing and touch—have hardly begun to be realized.

### Input Devices

Keyboards and mice now provide the means for almost all input. Despite great optimism over many years, effective technology for general-purpose speech and handwriting input remains elusive. Some observers question the desirability of these modes for many purposes. Speech is frequently imprecise and can be hard to edit. After learning to type, many people prefer keyboarding to (the originally even harder-to-learn method of) handwriting or dictation.

The question of which input modes are best for what tasks and which people is another issue much in need of research. One would not want to control a violin or a car by either typing or speech. What about all the other jobs in which a computer might eventually intercede between people and their work? Speech has been of great interest as a possible input mode for a long time. Progress, from some perspectives, has been slow; but the problems have been very difficult, and, although they need not all be completely solved before the technology can be applied effectively, sufficient progress must be made to get them over the threshold of usefulness. Speech recognition systems with limited ability are now on the market, and the technology is improving. How effective speech will be as an input mode, and under what circumstances it will be preferred to other possibilities, if and when the technology becomes sufficiently robust to support extensive use, remain to be seen. Much will depend on the degree to which the introduction of speech is guided by careful studies of users' needs and preferences.

Work on other modes—use of pointing devices other than the mouse, grasping of virtual objects, eye-fixation tracking—continues apace. A more mundane concern, but one of immediate practical interest is the need for a solution to the conflicting demands of mouse and keyboard.

### Intelligent Interfaces

The development of intelligent system interfaces is part of the plan of the U. S. government's 1993 High Performance Computing and Communications (HPCC) initiative. "In the future, high level user interfaces will bridge the gap between users and the NII. A large collection of advanced human/machine interfaces must be developed to satisfy the vast range of preferences, abilities, and disabilities that affect how users interact with the NII. Intelligent interfaces will include elements of computer vision and image understanding; understanding of language, speech, handwriting, and printed text; knowledge-based processing; and multimedia computing and visualization. To enhance their functionality and ease of use, interfaces will access models of both the underlying infrastructures and the users" (OSTP, 1994, p. 52). The interest in variety in interfaces stems, one assumes, from the intention that the NII be a resource that is available to, and approachable by, essentially the entire population.

## 1.5.8 Augmentation

Among the possibilities for more fundamental and dramatic advances we find the prospect of new tools to augment human intellectual and creative processes especially exciting. Consider three examples: math, writing, and art and music.

### Math

Tools for mathematical problem solving are of two kinds: those that facilitate the application of mathematical principles and methods, as appropriate, to the problems to be solved, and mechanical aids to computation. Problems can be difficult either for conceptual reasons or because of computational demands, or because of the way the two types of factors are intertwined. Suppose you want to solve a compound-interest problem; specifically, you want to know the monthly payments on a $112,000 30-year mortgage at 7.25 percent. There is a conceptually simple method for working the problem with pencil and paper: propose a trial monthly payment, calculate the interest for the first month, subtract it from the monthly payment and the result from the outstanding balance, and repeat the process for 360 payments, when you will find the error in your trial payment. Then pick a new payment value that will make a change in the right direction, say by splitting the difference between bracketing errors, until you get it right. Although this method is correct and foolproof, it obviously is very tedious and the mechanics are highly error prone.

Calculus provides a much less computationally intensive method for solving such problems, but the conceptually difficult principles of calculus elude the majority of people who pay mortgages (even if they have learned them in high school and applied them to just such problems). So they are left without a usable intellectual tool.

It seems at least plausible that most people would find the iterative trial and error method easier to grasp and apply because the correspondence between the problem and the solution method is more direct (see Nathan, Kintsch, and Young, 1992). It turns out that most spreadsheet programs contain methods to set up and execute iterative hill climbing methods and compute the answers quickly and accurately. Suppose high school students were taught these methods, and the HCI of the spreadsheet methods were improved, might this, perhaps, cause a leap forward in the practical intellectual competence of the average person, just as the provision of calculators can enhance the ability of people to do arithmetic?

We should note too that computers are being used by mathematicians today in the doing of pure, as well as applied, mathematics (Appel and Haken, 1977). Software now exists that can facilitate the doing of mathematics by mathematicians at all levels of competence, and some of it has the potential to be very useful in the teaching and learning of mathematics as well.

### Writing

Currently available computer tools for writing augment primarily the superficial mechanics of a creative process performed as always; they help with typographic editing, formatting and layout of documents, facilitate the organization, alphabetization and formatting of footnotes and references, give some assistance with spelling and a little with vocabulary and grammar. The more important aspects of composition and expression—generating information and ideas, creating clear and compelling arguments, composing efficient and effective narrative, or evocative and beautiful description, and ensuring discursive coherence and grace—are almost untouched.

There are promising avenues to be pursued in all these directions. Computer systems could help to find examples to quote, paraphrase or be inspired by, provide word and expression-finding aids more advanced than simple synonym lists and thesauri (e.g. type or mouse-select a phrase and get back words with related meanings and paragraphs from great literature or sections from textbooks or encyclopedias), evaluate and suggest improvements in coherence.

**Art and Music**

Most children love to draw until they "find out" that they are not good at it. Many adults like to invent tunes, so long as there is no danger of anyone else hearing them. Computer-aided tools for drawing, composition and performance are available that can be effective in the hands of experts (the first fully animated film, Toy Story, required years of effort by expert programmers and artists as well as enormous computing power and software sophistication); but the facilities currently available to the average computer owner are still crude toys, offering primarily mechanics for doing art or music in ways that mimic how it is done without the computer's help. New methods of creation and composition are needed that can make it easier for the average person to take advantage of new computer-based execution to produce things of satisfying originality and beauty.

So far, art creation systems have received very little user-centered design attention, almost no systematic analysis of what it is that ordinary people would want to do, find too hard to lead to satisfaction, and what would help them. When it has happened, we might, should we want to, all be able to draw lifelike portraits of our loved ones, animations and cartoons of our pets, produce good quality home movies, compose songs, set them to music and sing or play them with feeling and vivacity. We believe that the primary impediment to the realization of this vision is not technological limitations but a lack of understanding of users' needs and preferences with respect to artistic expression, and of the ways in which the technology presents the potential to meet them—e.g. to what extent it is good to make things easy, where does the optimum lie, for different folk, between a better color pallette and "paint-by-numbers?"

What will deeper HCI advances require? A primary need is the invention of new functions based on effective analyses of the cognitive processes involved in doing math, writing, and creating art, analyses that would reveal which parts of what processes are difficult and which are easy, which are desirable to make easier—or, perhaps more difficult—for whom and when, what alternative techniques could be enabled with new computer-aided mechanics and tools. Second, newly invented "cognitive tools" need to be implemented and then subjected to authentic formative evaluation involving actual trials with real people doing real work or play, trials from which what works and gratifies and what does not can be determined, and from which insights for better ideas can be gleaned.

The goal of all this, the hope for the future, is for computer systems to make it possible for ordinary people to perform intellectual and creative tasks at a markedly higher level than they can now, and the reason for that is to provide, through the interactive possibilities of computer mediation and augmentation, not only more effective and productive work tools, but, equally important, more active and productive avocational and recreational activities, activities that are at once more fulfilling and more growth-enhancing than the passive entertainment of television or the superficial excitement of current computer games.

The resources that already are available to the user of the Internet can represent a bewildering plethora of possibilities. To exploit the opportunities that are there, users, especially new users, need help. Steinberg (1994) identifies several tools that have been developed to help users find their way around the Internet and provides instructions as to how to avail oneself of each of them. The tool mix is changing rapidly, however, and we can expect it to continue to do so for the foreseeable future. As of this writing, nothing has become adopted by a sufficiently large segment of the user community to be considered an industry standard. Studies aimed at determining the kinds of help that users most need can contribute useful information to the process, as can studies of the tools that have been developed and efforts to discover ways that their effectiveness can be improved.

### 1.5.9 Information Finding, Use, and Management

Finding information is a generic problem that we all face, more or less constantly. One wants to discover what job opportunities exist for a specific vocation in a given geographical area, what houses are available for purchase in a particular town and within a specific price range, what universities have first-rate programs in applied mathematics, what is known about a new medication that one has been advised to try to combat a stubborn allergy.... At a more day-to-day level, one might want to know what specials the local supermarket is running this week, where to find a replacement for a worn-out part to a household appliance, how to fix a leaky faucet....

When information of the type that people often want is stored electronically, what is needed are effective search programs that can quickly find within the electronic databases just the information one seeks. This is a difficult problem in part because distinguishing between what is within the category of interest and what is not is typically a nontrivial task to automate. However, systems and tools that succeed in making it much easier for people to find the kinds of information that they often spend significant amounts of time seeking are clearly desirable objectives for research and development.

Harnad (1990) has noted the capability that electronic mail provides for researchers to distribute findings or prepublication drafts of scientific articles to individuals, groups, or even an entire discipline for comment and interactive feedback, and has referred to this medium as "scholarly sky writing." In Harnad's view, "sky writing offers the possibility of accelerating scholarly communication to something closer to the speed of thought while adding a globally active dimension that makes the medium radically different from any other" (p. 344). Levinson (1988) sees teleconferencing technology enabling what he refers to as "the globalization of intellectual circles." The idea of such circles is contrasted with Marshall McLuhan's "global village," the inhabitants of which receive common information from television but usually are not able to create or exchange it. "The computer conferencing network establishes on a geographically irrelevant basis the possibilities for reception and initiation and exchange of information that heretofore have existed only in in-person, localized situations. The result is that members of these electronic communities know each other much as members of a real-life local intellectual community such as a university do, but as members of a global television audience do not" (p. 205).

Complementary to the idea of sky writing is that of casting "does anybody know ..." questions on a network. Electronic bulletin board readers find such questions on many subjects addressed to the community at large in the hope that someone will have an answer and be willing to share it. One small study of the use of a corporate electronic bulletin board found that roughly one-quarter of 1000 messages posted during a 38-day period were either requests for information or posted replies to those requests (Nickerson, 1994). (Replies that went directly to the requestors instead of being posted on the bulletin board were not counted.) Requests for information covered a wide range of topics: Where can I rent a hall for a wedding reception? Find costumes for a masquerade party? Buy firewood? How do I send a message to a particular network host? Supplement heat from an oil furnace in an old drafty house? Convert home movies to VCR tape? Who can recommend an agent/company from whom to obtain an automobile/homeowner's insurance policy? How does one initiate a small claims court case? Please recommend hotels, inns, B&Bs, campgrounds, hiking trails, . . , in New England, in Montreal, in Oaxaca. What are some good hiking possibilities within one-and-one-half hours of Boston, not more than five hours up and down, and not mobbed? What is the origin of the Olympic torch?

This study focused on the bulletin board used primarily—though not exclusively—by the employees of a single technology-oriented company of modest size. When casting "Does anybody know ...?" questions on a large network, it probably makes sense to post them on specific bulletin boards whose communities of readers would be considered most likely to include some who knew the answers. There are many anecdotal reports of people who do just this and often to good effect. The Internet provides users with access to Usenet, which can be thought of as a collection of special-interest bulletin boards or discussion groups. As of 1994, Usenet was posting about 40,000 messages a day (Steinberg, 1994).

Although there are many tools and resources that can help one find information through the Internet—directories, bulletin boards, World Wide Web searchers—learning how to use these resources effectively can be a nontrivial task. There are many instruction and reference manuals that provide useful guidance both to the beginner and to the occasional user, but real expertise is acquired only through experience. Useful advice and guidance can be gained from other more experienced users on line; the casting of "does anybody know ..." questions can be productive in this regard.

An expected effect of the continuing development of information technology is much greater availability of information to the average person. We assume that this is a good and desirable thing. But this benefit could bring with it some unanticipated problems. Without effective tools that will help people deal with it, the greatly increased access could complicate life. What can be done to help people handle information glut? What can help them to keep from becoming paralyzed by a surfeit of choices? Programs that can search large databases or that can filter datastreams for information of interest to particular users are one approach to this problem that is being tried.

There is considerable interest in the development of electronic "agents" (personal digital assistants, gophers, alter-egos...) that can continuously scan large databases on behalf of individuals, finding and organizing the information they would want to have. There is some concern, however, that consumers are likely to have unrealistic expectations regarding the types of services that personal agents will be able to perform in the foreseeable future, especially if encouraged to do so by industry hype. As an antidote to the building of such unrealistic expectations, Browning (1995) has suggested that whenever one sees the phrase "intelligent agent" one should substitute "trainable ant" because that is a more appropriate model of what is likely to be available at least in the short term. In any case, the general question of how to make very large collections of electronically accessible information really useful to the average person presents a wealth of research, development and evaluation opportunities.

### 1.5.10 Person-to-Person Communication

We note again that, although it was not the purpose for which computer networks were developed, electronic mail, broadly defined to include electronic bulletin boards, discussion groups, and similar interperson-communication facilities, has turned out to account for a surprisingly large fraction of the total traffic over computer networks. Programs to permit the sending and receiving of messages were initially written to facilitate communication among engineers working on the development of network technology, but the potential inherent in this means of communication soon became apparent and motivated the development of formal email software for more general use. Advantages of electronic mail over more traditional communication media were pointed out by early enthusiasts (Bair, 1978; Lederberg, 1978; Uhlig, 1977). Possible disadvantages, or problems that the new capability might cause, received less attention.

According to Ratan (1995) the total volume of mail delivered by the U.S. post office has increased by about 5 percent since 1988, but the volume of business-to-business mail has decreased by 33 percent during the same period. Much of the decrease is assumed to have been lost to fax machines. At the present time, fax machines are much more likely to be found in offices than in homes, but they could become much more common in homes in the future. If they do, it seems reasonable to expect that much person-to-person mail will go the way that business-to-business mail seems to be going. The fax machine itself may be a temporary thing, however; in time affordable computer systems could provide all the capabilities now provided by PCs, phones and fax machines, and some others as well. Many uses of fax involve printing a document from a computer, transmitting it in facsimile, then rekeying or scanning some or all of its information into another computer. Obviously email, especially when it becomes more nearly universally available, is a considerable improvement over this process.

What are the implications for human communication of the fact that with email and related means of computer-mediated communication the impression one forms of people is based on what and how they communicate with these media and not on how they sound or look? Are people more likely to reveal their "true personalities" with computer-mediated communication than in face-to-face encounters, or does the email venue encourage and facilitate masquerading and the presentation of false identities? How will the character and quality of the communication change when voice and video are standard options?

One of the more thought-provoking findings to date on the difference between electronically mediated and face-to-face communication is that social status is a less important determinant of behavior, perhaps because status differences are less apparent in electronic communication. The proportion of talk and influence of higher status individuals is less when communication is via electronic mail, for example, and impediments to easy participation in meetings—soft voice, physical unattractiveness—seem to be less important factors in electronically mediated communication (Sproull and Kiesler, 1991a, b). This is not to suggest, that electronically mediated communication is to be preferred to face-to-face communication in any general sense, but only to point out that it may have advantages under certain circumstances.

It may of course have certain disadvantages as well. Email does not provide, for example, the rich context of nonverbal cues that carry nuances of meaning and affect. Kiesler, Siegel and McGuire (1984) identify "(a) a paucity of social context information and (b) few widely shared norms governing its use" (p. 1126) as two of the more interesting characteristics of computer-mediated communication from a social psychological point of view.

Face-to-face communication appears to follow certain rules even though most of us are probably not sufficiently aware of many of them—even though we follow them—to be able to articulate them very clearly. These rules have to do with selecting forms of address that are appropriate to the relationship between the conversing parties, keeping the affective qualities of the conversation within acceptable bounds, using changes in eye-fixation to signal changes in speaker–listener roles, filling intervals where a pause in the conversation would be awkward, terminating a conversation in an inoffensive way, and so on. Most of the conversational rules that we follow were not invented, like the rules for chess, but evolved over time, and it is not clear that all of them are yet well understood, despite considerable research directed at discovering them. Some of the rules that govern face-to-face conversations will probably transfer to computer-mediated conversations, and some will not; moreover, it is to be expected that rules unique to the computer-mediated situation will evolve over time, and that they too will be discovered only by research.

The question of control will be an important one as the patterns of use of electronic bulletin boards continue to evolve. There appears to be a fairly strong preference among most users for a system to be completely uncontrolled, but there is also a recognition that the lack of constraint sometimes results in the degeneration of effective discussions into intemperate shouting matches or worse. This is not unique to electronically mediated discussions, of course, but we have had considerably longer to work out some commonly accepted rules of etiquette for face-to-face discussions; it is likely to take a while to get to a comparable place with respect to discussions that occur in cyberspace.

### 1.5.11 Computer-Mediated Communication and Group Behavior

How will the greater person-to-person access that computer-mediated communication provides across different levels of an organizational hierarchy affect organizational structures of the future? It is generally assumed that the communication that occurs via computer networks is more lateral than that that occurs face-to-face or with other media. Perhaps because, as already noted, status differences are less apparent, the fact that communicants do not occupy the same level within an organizational hierarchy has less influence on communication within a network context than elsewhere. In any case, the network medium seems to foster a more egalitarian attitude among users than other communication media, and especially than face-to-face situations (Kiesler, Siegel, and McGuire, 1984).

To the extent that these perceptions are true, they raise the question of whether networks will tend, in general, to stimulate and facilitate the formation of groups of people with common interests (values, goals, ideologies) or to inhibit and impede the same. The question also arises as to how groups formed via network communication will resemble, and how they will differ from, groups formed in more traditional ways. Will they be more or less cohesive? More or less effective in realizing their group goals? More or less easily influenced by nonmembers? More or less open to change?

People with email access probably number in the tens of millions today. What will it mean when that number increases to hundreds of millions, as it is may well do in the foreseeable future, or to billions, which is not out of the question. In principle, everyone with a telephone—100 million in the U. S. alone—could have email as well. What should we make of Smolowe's (1995) claim that "[a]n estimated 80 percent of all users [of the Internet] are looking for contact and commonality, companionship, and community?"

How will computer-mediated communication compare with mass media in terms of the effectiveness with which it can be used for purposes of mass persuasion? For the dissemination of propaganda? Will it prove to be a useful tool for demagogues, or will it weaken demagoguery, in general, by making it easier to expose faulty arguments and false claims? All of these questions are challenges to research and innovation.

## 1.6 Summary and Concluding Comments

In a somewhat cynical assessment of the computer revolution and the "utopian fantasies" about the future course of it, Hughes (1995) has warned that "We will look back on what is now claimed for the information superhighway and wonder how we ever psyched ourselves into believing all that bulldust about social fulfillment through interface and connectivity" (p. 76). The picture he paints of the course the revolution could take presents the technology that is symbolized by computers and interactive multimedia as a powerful means of removing ourselves from reality, turning our heads to mush, and entertaining ourselves to death—more or less continuing the worst aspects of television but much more effectively. "Interactivity may add much less to the sum of human knowledge than we think. But it will deluge us with entertainment, most of which will be utter schlock" (p. 77). The picture is not as implausible as one would like it to be, in our view. We hope it proves to be wrong, but only time will tell.

Another possibility is that of what Dertouzos (1991) has referred to as the "hopeful vision of a future built on an information infrastructure that will enrich our lives by relieving us of mundane tasks, by improving the ways we live, learn and work and by unlocking new personal and social freedoms" (p. 62). This exciting and optimistic vision was expressed in the lead article of a special issue of Scientific American on "Communications, Computers and Networks," Other contributors to the issue also expressed generally optimistic expectations regarding the future impact of information technology on our lives, but they also showed some sensitivity to the fact that such a happy future is not assured. Noting that the new technologies could widen the gap between rich and poor, inundate us with "infojunk," and increase the incidence of white-collar crime and violations of privacy, Dertouzos noted the importance of efforts on the part of designers and users of this technology to attempt to steer it in useful directions. We agree with this and believe that efforts in HCI can play a consequential role in that steering.

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*Handbook of Human-Computer Interaction*
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# Chapter 2

# Information Visualization

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## 2.1 Introduction

Computation provides the most plastic representational medium we have ever known. It can be employed to mimic successful mechanisms of earlier media but it also enables novel techniques that were not previously possible. Computationally-based information presentations promise to dramatically enrich our understandings as well as assist us in navigating and effectively exploiting rapidly growing and increasingly complex information collections. In this chapter we survey a sample of recent information visualization research.

Information visualization has a long history, dating to the earliest forms of symbolic representation, and can be approached from multiple perspectives, ranging across psychology, epistemology, graphic design, linguistics, and semiology to newer perspectives emerging from cognitive science. There are numerous introductory surveys of information visualization. Examples include popular books by (Tufte, 1990) practical work on data visualization (Keller and Keller, 1993), and work applied to specific fields such as statistics (Cleveland and McGill, 1988). Information visualization research has grown dramatically in the last few years. Well-designed visualizations can be tremendously helpful but are still very challenging to create. There is an increasing amount of work on automating the production of visualizations (Mackinlay, 1996) and on providing tools to assist in designing effective interactive information visualizations. The Sage and Visage work of Roth and his colleagues (Roth, Kolojejchick, Mattis, and Chuah, 1995) is particularly noteworthy.

Our goal in this chapter is not a comprehensive survey of information visualization but rather to provide a glimpse of current research and attempt to communicate the exciting potential of new dynamic representations. To accomplish this we profile selected recent work from our research group and others. We then step back from the details of specific projects to discuss what we see as the beginnings of a paradigm shift for thinking about information, one that starts to view information as being much more dynamic and reactive to the nature of our tasks, activities, and even relationships with others.

## 2.2 Lenses As Information Filters

Consider the series of screen snapshots in Figures 1-4. They depict movable lenses (Stone, Fishkin, and Bier, 1994) that combine arbitrarily-shaped regions with filtering operators. They can be moved over objects to dynamically change the views presented. Figure 1 depicts placement of a pair of lenses over a section of text. The upper lens highlights text with special properties. In this example, it has highlighted Middle and Right and thus has provided additional information, regions that can be selected with the mouse, that would

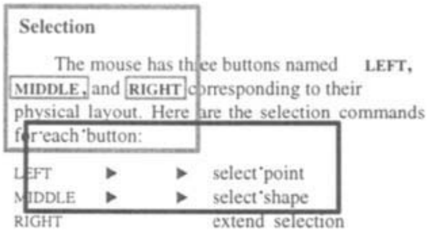


*Figure 1. **Pair of movable lenses.** Upper lens reveals special properties of text and the lower lens shows tabs and formatting.*

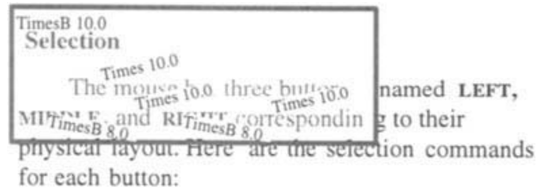


*Figure 2. **Font identification lens.** Shows fonts used in regions it is postponed over.*

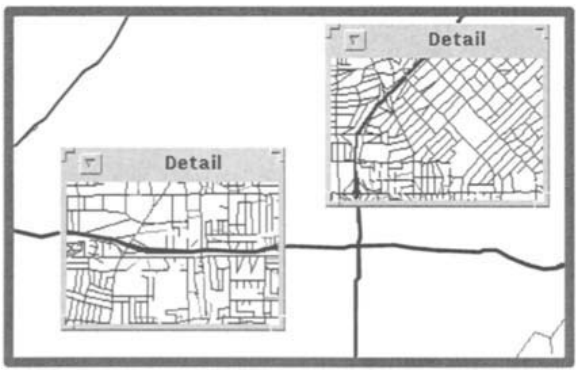


*Figure 3. **Detail lenses.** Here placed over the starting and ending location of a journey to assist in planning.*

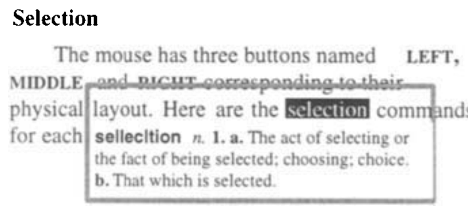


*Figure 4. **Definition lens.** Clicking through this lens displays the definition of a word.*

normally be invisible. The lower lens similarly shows whitespace and formatting codes such as tabs (here indicated by triangles).

The lens in Figure 2 allows the identification of fonts used in a portion of text. Figure 3 uses lenses to provide greater detail about sections of a map. In this particular example, lenses have been placed over the starting and ending locations of a journey to assist with planning. One, for example, typically needs much more detail about streets in the areas around the beginning and end of a journey. Figure 4 demonstrates a lens that one can click through to display the definition of a word.

These examples of simple interactive information visualization techniques employ a metaphor based on physical lenses but allow for creation of lenses with dynamic functionality that goes beyond mere magnification. They are movable regions that provide alternative representations of objects within an overall context.

There are a number of research groups exploring the use of lens-like interfaces to support information visualization. The work described above comes from researchers at the Xerox Palo Alto Research Center (Bier, Stone, Pier, Buxton, and DeRose, 1993). Earlier work at the MCC Human Interface Laboratory investigated similar interface mechanisms for selectively viewing information in large knowledge bases (Hollan, Rich, Hill, Wroblewski, Wilner, Wittenburg and Grudin, 1991). This technique, also termed lenses, provided for access to alternative perspectives on knowledge base entries. Related work on *brushing* to link data across different views has been very valuable for analyzing multi-dimensional data (See Cleveland and McGill, 1988). For the last several years we have also been exploring lens-like filters as part of the Pad++ dynamic multi-scale interface research effort that we describe next.

## 2.3 Pad++: A Zoomable Graphical Sketchpad

Imagine a computer screen made of a sheet of a miraculous new material that is stretchable like rubber, but continues to display a crisp image, no matter what the sheet's size. Imagine further that vast quantities of information are represented on the sheet, organized at different places and sizes. Everything you do on the computer is on this sheet. To access a piece of information you just stretch to the right part, and there it is. If you need more work room in some location, you just stretch the sheet to provide it.

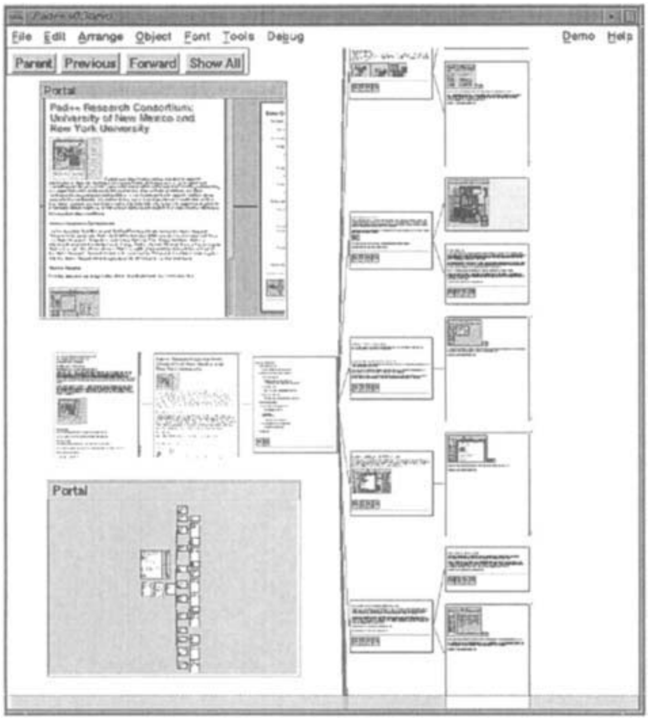

***Figure 5. Pad++ Web Browsing.*** *This depicts use of Pad++ for browsing the World Wide Web. Here we are looking at the Pad++ home page (www.cs.unm.edu/pad++). There are two portals providing views onto the same surface. In the top portal we have zoomed into the Pad++ home page. The lower portal shows a zoomed out view that allows us to see all the pages that have been traversed.*

In addition, special lenses come with this sheet that let you look onto remote locations. With these lenses, you can see and interact with many different pieces of data that would ordinarily be quite far apart. The lenses can also show different representations of the same underlying data and can even filter the data so that, for example, only data relevant to a particular task appears.

Imagine also new mechanisms that provide alternatives to scaling objects purely geometrically. For example, instead of representing a page of text so small that it is unreadable, an abstraction of the text, perhaps just a title that is readable, can be presented. Similarly, when stretching a spreadsheet, instead of showing huge numbers, it might be more effective to show the computations from which the numbers were derived or a history of interaction with them.

Pad++ (Bederson, Hollan, Perlin, Meyer, Bacon, and Furnas, 1996) provides the beginnings of an interface like this. We don't really stretch a huge rubber-like sheet, we simulate this with panning and zooming. By tapping into people's natural spatial abilities, Pad++ tries to increase users' intuitive access to information. Figure 5 depicts a view of the World Wide Web. Pad++ uses *portals* to provide lens-like features and also to support scaling data in non-geometric ways that we term *semantic zooming* or *context-sensitive rendering*.

One simple example of lenses in Pad++ is to support alternative views of numerical data. For certain tasks numerical data may be more effectively viewed as a scatter plot, bar chart, or some other graphic. The advantage of lenses is that they allow users to dynamically choose between alternative views of the information. By selecting a lens, users can chose a particular view of the data that best supports their current task. As Cleveland and his colleagues have demonstrated, some graphical depictions are better for supporting certain types of data comparisons. Lenses provide a technology that empowers users to participate in this representational choice. In addition, lenses make such displays available *when* and *where* needed rather than clutter the display with multiple representations.

As shown in Figure 6, we can also use lenses to change the behavior of objects seen through them, allowing design at a more abstract level. For example, we have implemented a pair of data entry lenses that can change a numeric depiction into a slider or dial, as the user prefers. By default, the entry mechanism allows input by typing. However, dragging the slider lens over a numeric entry changes the representation from

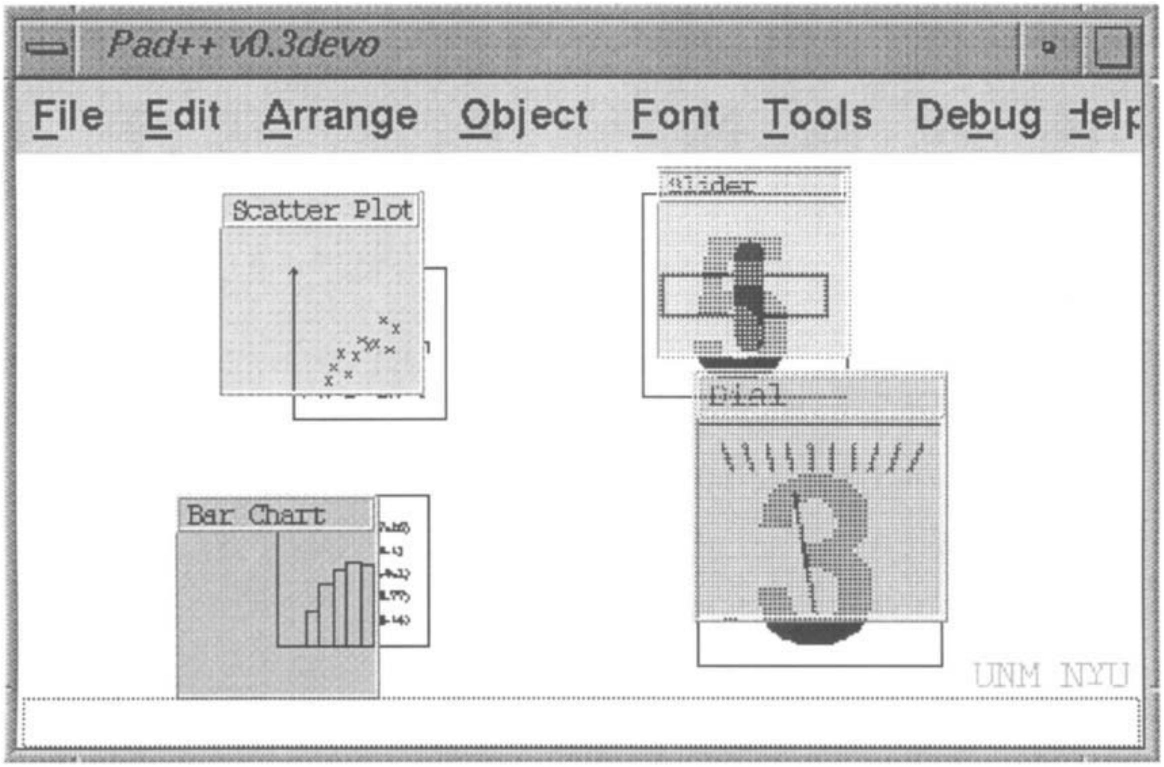


***Figure 6. Example Pad++ lens.*** *The lenses on the left depict numerical data as a bar chart or as a graph. The examples on the right demonstrate how lenses can be used to change methods of interaction. They allow numeric data to be changed using either slider or dial mechanisms.*

text to a slider, and then the mouse can be used to change the value by moving the slider. Another lens shows the data using a dial and permits modification by manipulating the dial's needle.

While it is effective to use lenses to see data in place, it is also sometimes useful to see different representations of data simultaneously and to be able to position these views at arbitrary locations. Portals are graphical objects that, in addition to acting as lenses, can provide a view of any portion of the Pad++ surface. Although a lens is used by dragging it over objects, a portal is used by panning or zooming within it to change where it is looking.

Easily finding information on the Pad++ surface is obviously very important since intuitive navigation through large dataspaces is one of its primary motivations. Pad++ supports visual searching with direct manipulation panning and zooming in addition to traditional mechanisms, such as content-based search. Some applications animate the view to the found items. These animations interpolate in pan and zoom to bring the view to the specified location. If the end point is further than one screen width away from the starting point, the animation zooms out to a point midway between the starting and ending points, far enough out so that both points are visible. The animation then smoothly zooms in to the destination. This gives a sense of context to the viewer, helps maintain object permanence, and speeds up the animation since most of the panning is performed when zoomed out.

We are exploring several different types of interactive visualizations within Pad++, some of which are described elsewhere (Bederson, Hollan, Perlin, Meyer, Bacon, and Furnas, 1996). Here we give one example: using Pad++ to develop a browser and navigation aid for the World Wide Web. It takes advantage of the variable resolution available for both representation and interaction. Layout of graphical objects within a multi-scale space is an interesting issue, and is quite different than traditional fixed-resolution layout. Deciding how to visually represent an arbitrary graph on a non-zoomable surface is extremely difficult. Often it is impossible to position all objects near logically related objects. Also, representing the links between objects often requires overlapping or crossing edges. Even laying out a tree is difficult because, generally speaking, there are an exponential number of children to fit in a fixed size space.

Traditional layout techniques use sophisticated it-

***Figure 7. Montage.*** *Montage application (on right) with regular browser (in back).*

erative adaptive algorithms for laying out general graphs, and can still result in graphs that are hard to understand. Large trees are often represented hierarchically with one sub-tree depicted by a single box that references another tree. Using an interactive zoomable surface, however, allows very different methods of visually representing large data structures. The fact that there is always more room to place information gives more options. Of course there are trade-offs involved in these representational decisions too. Pad++ is particularly well suited for visualizing hierarchical data because information that is deeper in the hierarchy can be made smaller. Accessing this information is accomplished by zooming.

In traditional window-based systems, there is no graphical depiction of the relationship among windows even when there is a strong semantic relationship. For example, in many hypertext systems, clicking on a hyperlink brings up a new window with the linked text (or alternatively replaces the contents of the existing window). While there is an important relationship between this linked information (parent and child), this relationship is not represented.

We are experimenting with a variety of multi-scale layouts of hypertext document traversals in which relationships between linked pages are represented visually. In one experimental viewer when a hyperlink is selected, the child document is shown to the side and smaller, and the view is animated to center the new object. After many links have been followed, zooming out gives an overview of the data, and the relationships between links is retained in the layout. In another viewer, tree relationships are explicitly depicted by edges between the pages of an html document. An alternative viewer provides the user with a camera that can be moved around the tree and the nodes it is looking at are show via a Pad++ portal.

We have also constructed facilities so that tree visualizations of web traversal can be pruned and annotated. Currently we are adding the ability for such trees to be checked periodically to see if the information they point to has changed. Annotations can be added to the tree to indicate changed sections. We are also investigating animations to take a user to places on pages they have saved that have changed since the last time they visited. This type of history-based information is discussed further in the section on history-enriched digital objects. We now turn to an information visualization system that provides image montages to assist in finding relevant and interesting web pages.

## 2.4 Montage

Web browsing is popular because hypertext is recognized as an easy way to access complex and interrelated information. But hypertext structures are often complex and their designers can't anticipate the particular needs and interests of every reader. Getting lost in hyper-space is all too common. Web browsers that only display one or two pages at a time don't provide enough context to help people stay oriented. The Pad++ web browser allows one to see many more pages and provides a focus plus context view so that one can see a page in the context of local traversals. Imagine instead, if there were an effective way to see hundreds of web pages at once.

Montage is a new way to surf the web by watching hundreds of images from web pages. Figure 7 shows a Montage application. Select an image that looks interesting to drive your regular web browser to the associated page. Montage is a fast way to visualize the landscape of the web because it utilizes the human capacity to recognize patterns in many images at once. Montage is effective because a large percentage of web images have content that is either explicitly textual or in some way relevant to the text on the associated web page.

Montage changes the nature of web navigation, avoiding some of the problems associated with hypertext navigation. By removing images from their associated web pages, Montage flattens the hypertext structure of the web. Similar images begin to form relationships, clusters, and streams. Relationships among images may only exist in the user's mind or the images may, in fact, have come from related pages. It hardly matters. Since hypertext structures are usually designed by someone who couldn't anticipate the particular needs and interests of all users, the actual hypertext structure is far less relevant than the relationships imposed on the information by a user through their own process of searching and browsing.

Before Montage can display images, it must first determine a mapping of images to associated web pages. A mapping of images to web pages can be obtained from many sources. In one version of Montage, the mapping is obtained from a local proxy server log file (and the images are obtained directly from the proxy server cache). Caches of HTML pages and images are meant to minimize access delays, but Montage reveals the cache as a community-wide resource. Images in the cache illustrate how a community uses the web and what subset of the web they find important. Montage monitors the cache, displaying any new images as they appear. When images are ordered by access time, streams of similar images identify active hypertext trails. Several images of new cell phones, for example, probably indicate that someone in your community is searching for information about purchasing a new cell phone. If this is of interest, select a cell phone to follow that hypertext trail.

It is also possible to obtain a mapping of images to associated web pages by parsing HTML pages and searching for embedded image tags. Although parsing HTML is not as fast as the proxy-server approach, it may be more generally useful. For example, imagine submitting a query to your favorite search engine on the web, but instead of having to read through the typical textual representation of the web pages that match your query, you can sit back and watch a Montage of images taken from the web pages that match your query. It should be easier to pick out the relevant images in the Montage than it is to pick out the relevant URL's in a large page of text. As another example, you may want to see a Montage of images taken from the pages listed in your hotlist or bookmarks file that have changed since you last saw them.

A snapshot of any portion of a Montage may be saved, annotated, and shared with others on the web. Montage snapshots are stored as imagemaps. Imagemaps, which have become a de-facto standard for implementing graphical menus on the web, are single images with associated coordinate files that maps regions of an image to a URL. When a user selects an imagemap, the coordinates of the selection point are sent to the web server, which searches through the associated coordinate file, determines which region was selected, and drives the user's web browser to the associated URL.

A Montage snapshot is a composite of all the images that were visible in the original Montage when the snapshot was created. The coordinates of each Montage image and the URL of its associated web page are stored in the imagemap coordinate file. Although Montage snapshots are static, when installed on a web server as an imagemap they retain the interactive quality of the original Montage. Select any region of the imagemap to drive your browser to the associated web page, just as if you had selected the image in Montage.

It's easy to add annotations to snapshots. Since imagemaps are designed to appear on web pages with text, annotations take the form of arbitrary HTML. Annotated snapshots seem particularly useful when the Montage images are ordered by internet site. Snapshots of images from the same site form a visual preview, or Postcard, of the site. Postcards are useful for illustrating bookmarks and hotlists. Annotated Postcards are useful for sharing information about web sites. An example of one particularly useful application is a shell script that automatically illustrates "surf reports" by converting email site reviews into web pages with interactive Postcards.

When Montage installs a Postcard on a web server it gives it a unique URL. To share the Postcard with a friend, send them the Postcard's URL via email. When they drive their browser to the URL they can see your snapshot and read your annotations. If your friend wants to visit the actual web site all they have to do is select an image on the snapshot. Surely something one can't do with a cardboard postcard!

Perhaps the most important idea embodied by Montage is that it shows people what is available on the web and it makes it easy to access web pages about things that seem interesting or relevant. We call this sort of activity "Passive Surfing" because it makes web surfing dramatically easier than clicking through irrelevant hypertext structures. But while Montage focuses exclusively on images, there are in fact many other types of multimedia files on the web: audio, MIDI, video, animation, etc. Like images, many of these multimedia files are associated with textual web pages that contain associated information. Future systems for Passive Surfing will play audio and MIDI files while displaying video and animation files and continuing to make it easy to access the associated web pages for things that seem interesting or relevant.

But why limit ourselves to the web? Even without the web, our computers are typically loaded with information that that is difficult to use for two major rea-

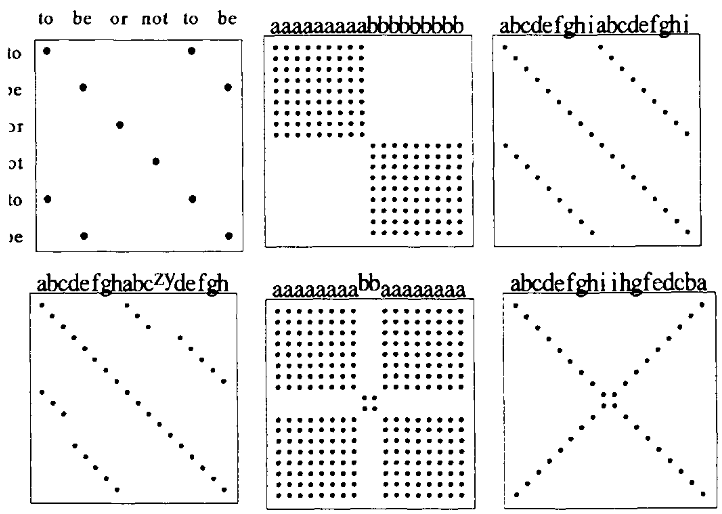


***Figure 8. Dotplot.*** *a)Six words of Shakespeare, b)Squares, c)Diagonals, d)Broken Diagonals, e) Light Cross, f) Palindrome.*

sons: 1) we don't know what's available, and 2) we don't know how to access it. There seems to be a pressing need for a general class of systems for *Passive Browsing*: a lazy way to access information that uses animation to illustrate what is available and which actions are possible. Like trailers for motion pictures and demo modes for arcade games, a Passive Browser entices with snippets of representative action. Unlike trailers and demos, however, a passive browser affords user interaction. Users can pause or replay the animations and select images to obtain additional information. Each user's selections can be collated to build a profile of that user's interests, which can in turn be used to guide the creation of the animations. A passive browser might take the form of an interactive screen saver that illustrates available on-line services and responds to gestures by providing additional information or launching applications. Most screen savers cause our computer screens to either display no images or to display completely gratuitous graphics. Instead, why not seize this opportunity to show people what sorts of information is available, while at the same time making it easy for them to access the information that they find interesting and relevant?

Montage exploits our ability to see patterns in information. A Montage of a web site provides a graphical overview of the available information. We turn now to a novel information visualization technique specifically designed to assist us in seeing patterns and structure in large quantities of information.

## 2.5 Dotplot

Dotplot is a technique for visualizing patterns of string matches in millions of lines of data, text, or code. Dotplots were originally used in biology to study similarities in genetic sequences (Maizel and Lenk, 1981; Pustell and Kafatos, 1982). Since then, the Dotplot technique has been extended and improved (Church and Helfman, 1993). Dotplot patterns may now be explored interactively, with a Dotplot Browser, or detected automatically with a variety of image processing techniques. Current applications include text analysis (author identification, plagiarism detection, translation alignment, etc.), software engineering (module and version identification, subroutine categorization, redundant code identification, etc.), and information retrieval (identification of similar records in results of queries).

Dotplot patterns reveal a hidden, unifying structure in the digital representations of information. Information is always encoded in a particular representation: programs and texts are stored in a particular language; data is stored in a particular format; DNA is represented by sequences of nucleotides, etc. Whatever the representation, streams of information can be tokenized and matching tokens can be plotted. The patterns that appear in the plots reveal universal similarity structures that exist in all digital objects at multiple scales and levels of representation.

The Dotplot technique is illustrated in Figure 8a. A sequence is tokenized and plotted from left to right and top to bottom with a dot where the tokens match. Dots off the main diagonal indicate similarities. While Figure 8a shows six words of Shakespeare, Figure 9a shows "The Complete Works". Grid lines show the boundaries between concatenated files. Dark areas show a high density of matches, while light areas show

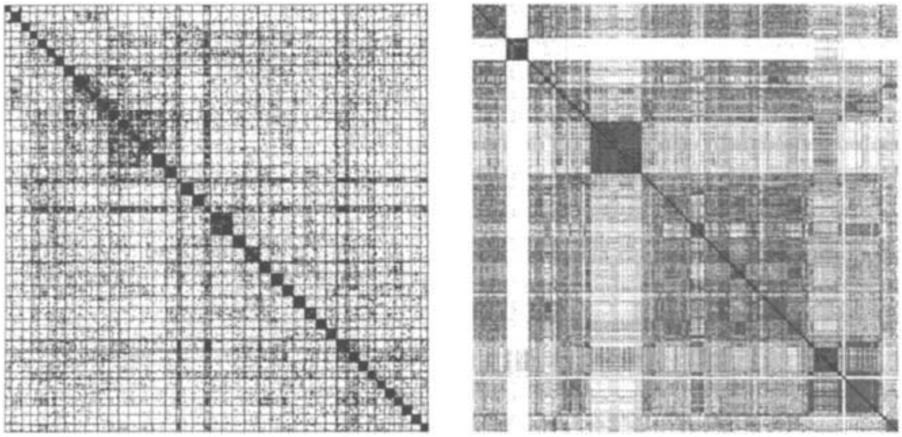

***Figure 9. Dotplot.** a) A million words of Shakespeare, b) About two million lines of C code: an entire module of telecommunications switching program.*

a low density of matches. Weighting and reconstruction methods are used to display matches from more than one pair of tokens in a single pixel. Weighting prevents matches between frequent tokens from saturating the plot. Additional details of the dotplot technique are described elsewhere (Church and Helfman, 1993).

Previous approaches to detecting similarity do not reveal the richness of the similarity structures that have been hidden in our software, data, literature, and languages. Information retrieval techniques that model documents as distributions of unordered words, can identify similar documents, but can not identify copies, versions or translations. Dynamic programming techniques (e.g. the UNIX *diff* utility) and algorithms that identify longest common substrings have difficulty dealing with noisy or reordered input. In contrast, the Dotplot technique is remarkably robust and insensitive to noise because it relies on the human visual system for identify patterns.

Dotplot patterns are interpreted though a visual language of squares and diagonals (Church and Helfman, 1993). Figures 8b-f show Dotplots of artificial character sequences that have been tokenized into characters and plotted with a dot where the characters match. Figure 8b shows that squares identify unordered matches (documents with lots of matching words or subroutines with lots of matching symbols), while Figure 8c shows that diagonals identify ordered matches (copies, versions, and translations). Although dotplots often appear complex, they can always be interpreted in terms of their constituent squares and diagonals.

The Dotplot technique scales up to very large data sets. No matter how many matches are plotted, the interpretation of squares and diagonals is still the same. For example, the small, dark squares along the main diagonal of Figure 9a are caused by the names of the casts of characters in each of Shakespeare's plays, which generally match within a single play, but not across different plays. There are also dark squares along the main diagonal of Figure 9b, a Dotplot of about two million lines of C code. The input to Figure 9b (and the other dotplots of C code, Figures 10 and 11a) has been tokenized into lines and plotted with a dot where entire lines of code match. The squares in Figure 9b identify submodules of a large telecommunications switching system; comments and statements generally match within a single module, but not across different modules. In both Figures 9a and 9b, squares identify unordered matches. As another example, in Figure 10a, a Dotplot of about 3000 lines of C code (from the same telecommunications switching system shown in Figure 9b), diagonals identify groups of several hundred consecutive lines of code that were copied at least four times. In Figure 10b, a Dotplot of about 67,000 lines of C code, the long, broken, diagonals identify different versions of the same program. In each case, diagonals identify ordered matches. Broken diagonals, shown synthesized in Figure 8d, identify insertions of unordered tokens into otherwise ordered matches. In Figure 10b, each discontinuity in the diagonals indicate lines of code that were added to the new version of the program.

The Dotplot technique reveals similarity structures at multiple levels of representation. For example, the grid-like texture of Figure 11a is formed by repeated "break" tokens in a C program "switch" statement. This pattern is not due to a particular programmer, but rather to the designers of the C programming language itself, who chose to require an explicit syntactic struc-

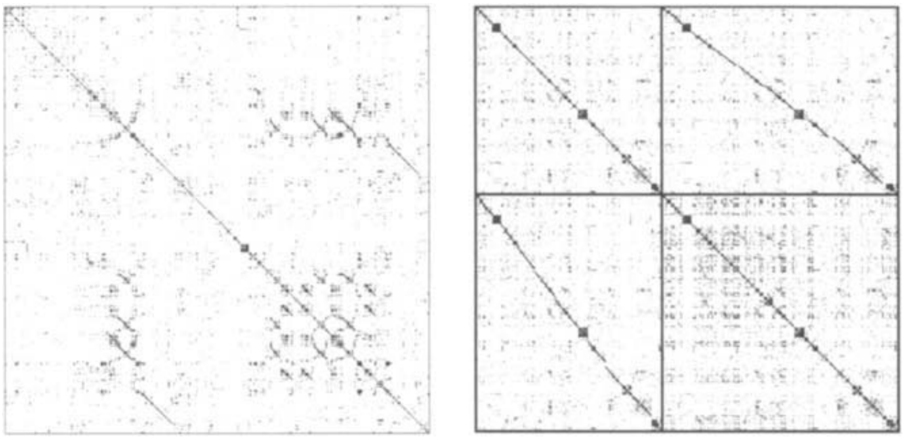

***Figure 10. Dotplot.*** *a)Copies in 3400 lines of C code, b)two versions of the X Toolkit (66,600 lines of C code).*

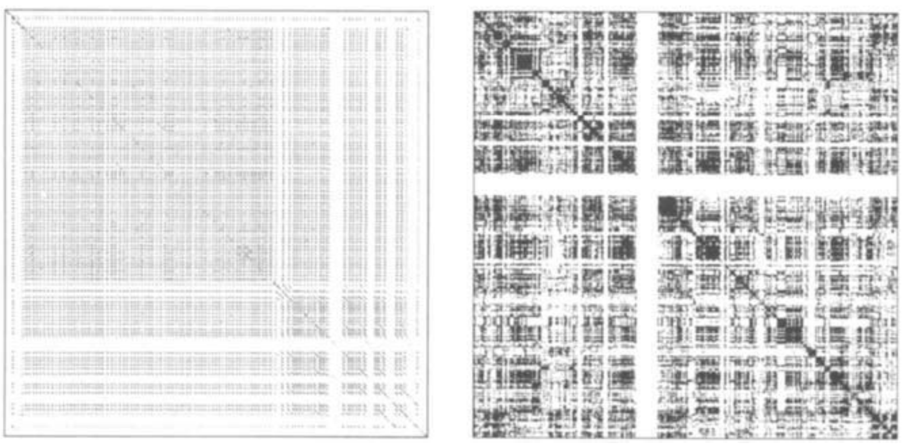

***Figure 11. Dotplot.*** *a) A 1000 line C switch statement, b) 290,000 file names.*

ture ("break") to achieve a commonly needed function (terminate a switch statement). As a result, grid-like textures occur frequently in Dotplots of C code.

The analysis of string matches might seem to be limited, static, and literal. If you analyze only matching strings in a program's code or data, you might expect to learn about the program's syntax, but you might not expect to learn anything about the program's semantics, how it works, runs, or is used. However, in some cases at least, the similarity structure of data related to a program does reveal aspects of how the program is used, and how it works. For example, Figure 11b is a Dotplot of several hundred thousand file names, each of the files on our laboratory's file server one particular afternoon. The file names appear in *inode* order (the order in which they were created) and a dot appears wherever two file names match.

The large white cross that spans Figure 11b is modeled by the character sequence of Figure 8e in which a sequence of *b*'s is inserted into a larger sequence of *a*'s. In general, light crosses indicate insertions into unordered sequences. The light cross in Figure 11b is caused by several thousand files with unique names that were created at about the same time. These file names were in fact generated by the UNIX *split* utility. Someone was in the process of sending the X Window System to Spain via email! Near the center of Figure 11b is a dark square, which when plotted with greater resolution (see Figure 12a), is shown to be formed by a pattern of equally-spaced diagonals. This pattern indicates that the file system is used to store many sets of files with exactly the same names, in exactly the same order---in fact, the file system is being used to store many different versions of the same, large program.

It is important to emphasize that both Figures 11b and 12a plot data associated with a particular program (i.e. file names). The code for the file system program

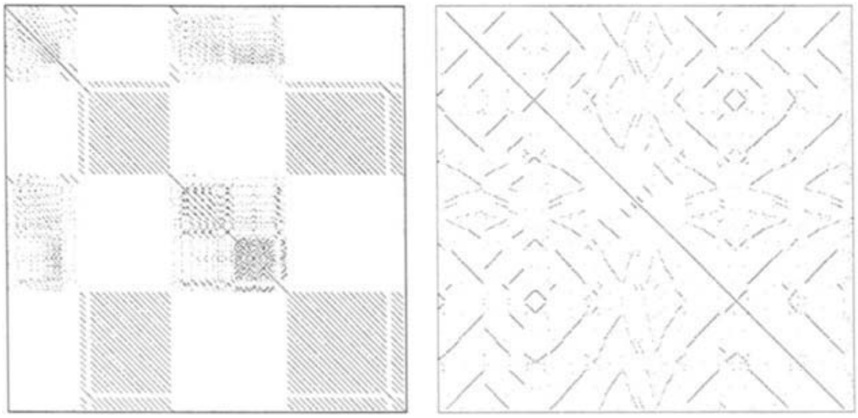

*Figure 12. Dotplot. a)Versioning in 8000 file names, b) Palindromes in 900 file names.*

itself is not plotted. Yet Figures 11b and 12a identify two different ways in which the file system program is used (as temporary storage while transferring large programs and as permanent storage for versions of large programs). An even more stunning example is shown in Figure 12b (a detail of Figure 12a). Figure 12b shows reverse diagonals, a rare form of diagonal that is hardly ever seen in dotplots of text or software (yet quite frequent in dotplots of music). A reverse diagonal, modeled in Figure 8f, indicates a palindrome, a sequence that is the same forwards as it is backwards. In Figure 12b the palindromes are formed by sequences of file names that appear first in one order, then later in the reverse order. These files were generated by the UNIX *make* utility that was used to quickly free several inodes ("make clean") and then reallocate them ("make"). The file system evidently stores inodes in a stack (i.e. a data structure with a Last-In-First-Out or LIFO access policy) so that they are reallocated in an order that is the reverse of the order in which they are freed. In this case, the Dotplot identifies a design decision in the file system code itself (which isn't even being plotted!).

Although understanding the similarity structure of large, complex Dotplots (such as the one shown in Figure 9b) is straightforward, it can be tedious. In an attempt to make large, complex Dotplots even easier to understand, we have begun experimenting with interactive Dotplot tours by combining Dotplots with Pad++. A Dotplot can be analyzed automatically and regions of the plot that contain high densities of matches can be precomputed at different resolutions along with textual representations of the input tokens most responsible for the matches. The precomputed plots can then be automatically scaled and positioned over an image of the original Dotplot in Pad++. Animations can be generated that zoom through the precomputed plots while updating a textual view of the associated matching tokens. At any time, users watching the animated tour can pause the animation to obtain additional information about the data set, compute additional plots, etc.

In summary, we have shown that although dotplots are primarily a technique for plotting literal string matches in static data, they are remarkably general, robust, resolution independent, representation independent, and can even reveal dynamic information about how programs are used and designed. In addition, we have described some preliminary experiments that promise to make Dotplots even more dynamic and interactive.

## 2.6 Information Visualizer

Card and colleagues have developed the *Information Visualizer* as part of an effort to attempt to better understand information workspaces. The system is based on the use of 3D/Rooms (Henderson and Card, 1986) for increasing the capacity of immediate storage available for the user, what they term the *Cognitive Coprocessor Scheduler-Based* user interface interaction architecture for coupling the user to information agents, and the use of visualization for interacting with information structure. (Card, Robertson, and Mackinlay, 1991).

The system is motivated by looking at information-based work in terms of cost structure. While it is widely appreciated that it costs to acquire, organize, and store information, the designers of the Information Visualizer emphasize that it also costs to access information and that a most important potential benefit of information visualization is that it can lower the cost structure of information access. For example, effective information visualizations often make aspects of in-

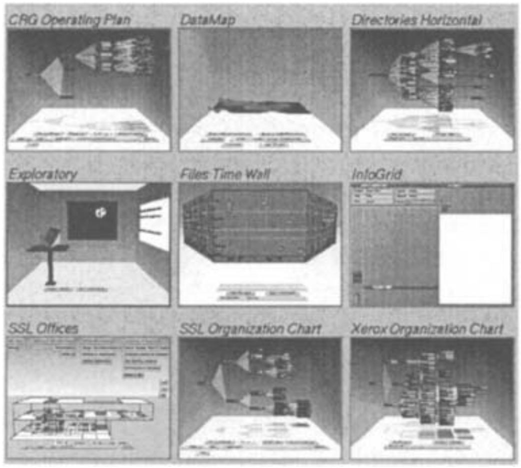


***Figure 13. Information Visualizer.** This depicts a variety of Information Visualizer applications. These include visualization techniques for browsing hierarchical structure (Cone Trees and Cam Trees) and linear structure (Perspective Wall). See Card, Robertson, and Mackinlay for more detail about these and other Information visualizer applications.*

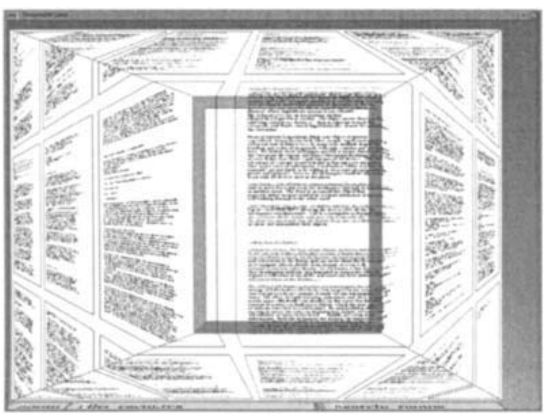

***Figure 14. Document Lens.** This example of a document lens uses a rectangular lens with elastic sides that produces a truncated pyramid. The sides of the pyramid contain the portions of the document not visible in the rectangular area. This is another example of use of the focus plus context information visualization technique first explored in Furnas work on fisheye lenses.*

formation that would normally have required inference directly perceptually available. Turning what would have required inference into direct perception is an example of changing the cost structure of information.

Card, Robertson, and Mackinlay make an analogy to the cost structure of sources of information for a typical office worker. Such an office worker can access a variety of information sources ranging from a computer terminal to files in a filing cabinet or books in a bookcase. They argue that each piece of this information has a cost structure associated with finding and accessing it and point out that what is normally meant by an "organized office" is one arranged so as to have a low cost structure.

Figure 13 gives a schematic overview of Information Visualizer projects. Main features of this work are the use of 3D perspective to provide focus plus context views of information and the use of double-buffering and other animation techniques to provide smooth interaction. Of central importance is a cognitive coprocessor architecture. This architecture is designed to increase the information interaction coupling with users. The architecture is based on a scheduler that continuously tries to support a level of interactivity appropriate for particular taks. The system will, for example, degrade rendering if that is needed to be able to maintain an appropriate level of responsiveness.

Figure 14 depicts lenses used with the Information Visualizer. In this case it is a document lens, a rectangular lens with elastic sides, that provides a focus plus context view of a document. Figure 15 shows two alternative techniques for depicting hierarchical structure. The figure at the top shows a cone tree visualization of a hierarchical directory structure. The visualization at the bottom shows an alternative layout. Again, notice the use of 3D perspective to provide fisheye views. Also when a node is selected the tree smoothly rotates so that nodes on a path to the root node are brought to the front. Animation is at a rate such that object constancy is maintained and thus the cognitive cost for the user of reorienting to the new layout is minimized.

## 2.7 History-Enriched Digital Objects

In addition to visualizing the information structure inherent in objects, we can also visualize the history of interaction with digital objects. By recording the interaction events associated with use of objects (e.g. reports, forms, source-code, manual pages, email, spreadsheets) makes it possible on future occasions, when the objects are used again, to display graphical abstractions of the accrued histories as parts of the objects themselves. For example, we depict on source code its copy history so that a developer can see that a particular section of code has been copied and perhaps be led to correct a bug not only in the piece of code being viewed but also in the code from which it was derived.

The basic idea behind history-enriched digital ob-

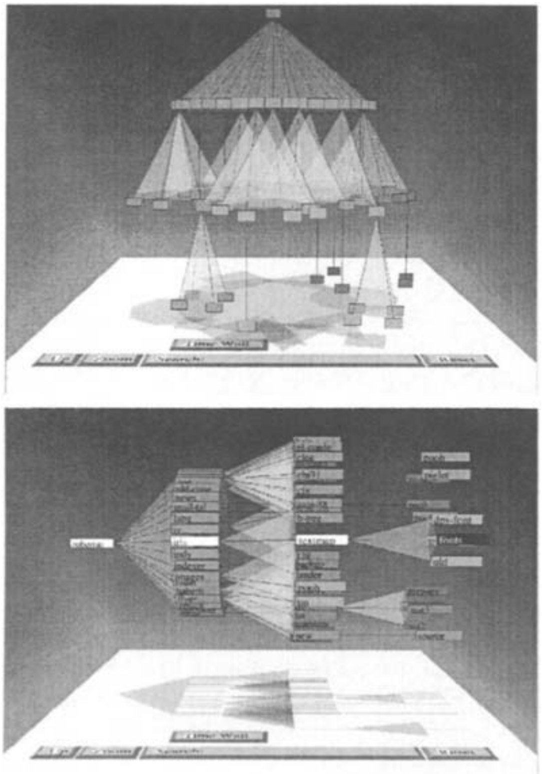

Figure 15. Cone Trees. Use of three dimensions to view hierarchies. At the top is a simple cone tree. When a note is selected the tree rotates so that nodes on the path to the top node are brought to the front.. Animation is at a rate that object constancy is maintained. The lower figure shows an alternative layout.

jects is to record the history of use on interface objects. We maintain a history of interactions and then attempt to display useful graphical abstractions of that history as parts of the objects themselves. In our interaction with objects in the world there are occasions when the history of their use is available to us in ways that support our interaction with them. For example, the well worn section of a door handle suggests where to grasp it. The binding of a new paper back book is such that it opens to the place we stopped reading last time. The most recently used pieces of paper occupy the tops of piles on our desk. The physics of the world is such that at times the histories of our use of items are perceptually available to us in ways that inform and support the tasks we are doing. We can mimic some of these mechanisms in interface objects. Of more importance we can also look for new types of mechanisms we can exploit in a computational media that have similar informative and supportive properties.

One of our first efforts was to explore the use of attribute-mapped scroll bars as mechanisms to make the history of interaction with documents and source code available. We have modified various editors to maintain interaction histories. Among other things, we recorded who edited or read various sections of a document. The history of that interaction was then graphically made available in the scroll bar. Figure 16 shows a history of editing of a file of code. We can see which sections have been edited most recently and who has edited various sections. A feature of representing this in the scroll bar is that it makes good use of limited display real estate. To investigate any particular section, users need only point at click at that section of the scroll bar.

We have explored a variety of other applications of the idea of history-enriched digital objects. For example, one can apply the idea to menus so that one might see the accrued history of menu choices of other users of a system by making the more commonly used menu items brighter. Or one can present spreadsheets such that the history of changes of items are graphically available. Thus, one easily can see which parts of a budget were undergoing modification. We have also explored applications that record the time spent in various editor buffers to allow one to see a history of the amount of time spent on the tasks associated with those buffers. We also record the amount of time spent reading wire services, netnews, manual pages, and email messages. This type of information can then be shared to allow people to exploit the history of others interaction with these items. One can, for example, be directed to news stories that significant others have already spent considerable time reading or to individuals who have recently viewed a manual page that you are currently accessing.

Of course, there are complex privacy issues involved with recording and sharing this kind of data. In our view, such data should belong to users and it should be their decision what is recorded and how it might be shared.

## 2.8 Toward a New View of Information

We hope this brief survey of recent informational visualization research has expanded your view of the possibilities for new dynamic information entities. We think the future holds an ever richer world of computationally-based work materials arrayed in multiple spaces that exploit representations of tasks, semantic relationships explicit and implicit in information and in our interactions with it, and user-specified tailorings to

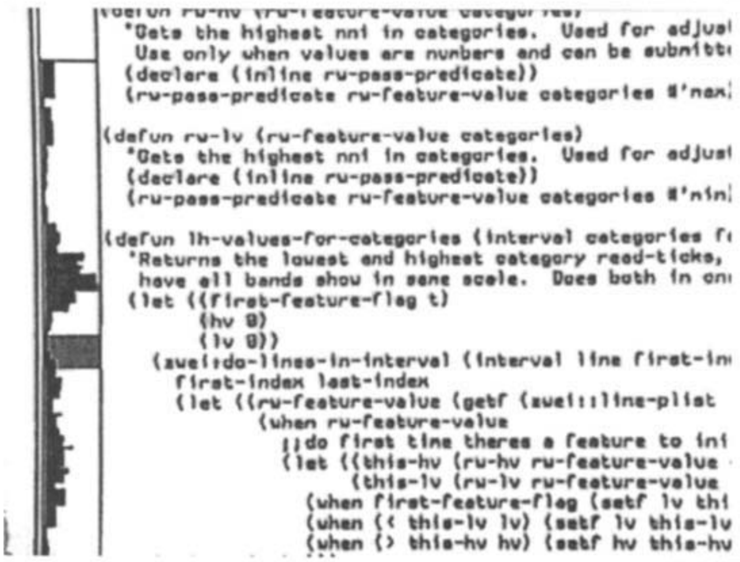


***Figure 16. Attributed-Mapped Scroll Bar.*** *The scroll bar depicts a graphical summary of the history of editing this code.*

provide effective, enjoyable, and beautiful places to work. Underlying this work is the beginnings of what we think is a paradigm shift for thinking about information, one that starts to view information as being much more dynamic and reactive to the nature of our tasks, activities, and even relationships with others.

In our view, this paradigm shift is evidenced in interfaces starting to move beyond mimicking mechanisms of earlier media to more fully exploit computationally-based mechanisms. While there is clearly much to be gained from continuing to explore traditional approaches, we see this new paradigm as expanding the space of interface possibilities. It also brings with it major challenges associated with developing new dynamic representations.

While interface designers have traditionally followed a metaphor-based approach, we see these new areas of the design space as arising from what we term an *informational physics* approach to interface design. Viewing information as being more dynamic and operating according to multiple alternative physics highlights a shift in focus to exploration of new representational mechanisms that do not necessarily have origins or analogues in the mechanisms of earlier static media. While an informational physics strategy for interface design certainly involves metaphor, we think there is much that is distinctive about a informational physics approach. Traditional metaphor-based approaches map at the level of high-level objects and functionality. They yield interfaces with objects such as windows, trash cans, and menus, and functions like opening and closing windows and choosing from menus. While there are ease-of-use benefits from such mappings, they also orient designers towards mimicking mechanisms of earlier media rather than towards exploring potentially more effective computer-based mechanisms. Semantic zooming is but one example mechanism that we think more naturally arises from adopting an informational physics strategy.

We are not the first to propose a physics-inspired view of interface design. It derives, like many interface ideas, from the seminal work of Sutherland on Sketchpad. Simulations and constraint-based systems that fostered the development of direct manipulation style interfaces are other examples of this general approach. They too derive from Sutherland and continue to inspire developments. Recent examples include the work of Borning and his students (Borning, 1979 and Borning, 1981). Witkin (Harada, Witkin, and Baraff, 1995) in particular has taken a *physics-as-interface* approach to construction of dynamic interactive interfaces.

Smith's Alternate Reality Kit (Smith, 1987 and Smith, 1996) and languages such as Self (Ungar and Smith, 1987) are also examples of following a physics-based strategy for interface design. These systems make use of techniques normally associated with simulation to help "blur the distinction between data and interface by unifying both simulation objects and interface objects as concrete objects" (Chang and Unger, 1993). More importantly, they are based on implementation of mechanisms at a different level of abstraction than is traditional. Smith, for example, gives users access to control of parameters of the underlying physics in his Alternate Reality Kit. With this approach comes the realization that one can do much more than just mimic reality. As Chang and Unger point out in their discussion of the use of cartoon animation mechanisms in Self, "adhering to what is possible in the physical world is not only limiting, but also less effective in achieving *realism*."

One aspect of our argument here is that it is important to look at the costs as well as the benefits of traditional metaphor-based strategies. We think it is crucial to not allow existing metaphors, even highly successful ones, to overly constrain exploration of alternative interface mechanisms. While acknowledging the importance and even inevitability of the use of metaphors in interface design, we feel it is important to understand that they have the potential to limit our views of possible interface mechanisms in at least four ways.

First, metaphors necessarily pre-exist their use. Pre-Copernicans could never have used the metaphor of the solar system for describing the atom. In designing interfaces, one is limited to the metaphorical resources at hand. In addition, the metaphorical reference must be familiar in order to work. An unfamiliar interface metaphor is functionally no metaphor at all. One can never design metaphors the way one can design self-consistent physical descriptions of appearance and behavior. Thus, as an interface design strategy, physics, in the sense described above, offers more design options than traditional metaphor-based approaches.

Second, metaphors are temporary bridging concepts. When they become ubiquitous, they die. In the same way that linguistic metaphors lose their metaphorical impact (e.g., *foot of the mountain* or *leg of table*), successful metaphors also wind up as dead metaphors (e.g. file, menu, window, desktop). The familiarity provided by the metaphor during earlier stages of use gives way to a familiarity with the interface due to actual experience.

Thus, after a while, it is the actual details of appearance and behavior (i.e. the physics) rather than any overarching metaphor that form much of the substantive knowledge of an experienced user. Any restrictions that are imposed on the behaviors of the entities of the interface to avoid violations of the initial metaphor are potential restrictions of functionality that may have been employed to better support the users' tasks and allow the interface to continue to evolve along with the users' increasing competency.

Similarly, the pervasiveness of dead metaphors such as files, menus, and windows may well restrict us from thinking about alternative organizations of interaction with the computer. There is a clash between the dead metaphor of a file and newer concepts of persistent distributed objects.

Third, since the sheer amount and complexity of information with which we need to interact continues to grow, we require interface design strategies that *scale*. A traditional metaphor-based strategy does not scale. A physics approach, on the other hand, scales to organize greater and greater complexity by uniform application of sets of simple laws. In contrast, the greater the complexity of the metaphorical reference, the less likely it is that any particular structural correspondence between metaphorical target and reference will be useful. We see this often as designers start to merge the functionality of separate applications to better serve the integrated nature of complex tasks. Metaphors that work well with the individual simple component applications typically do not smoothly integrate to support the more complex task.

Fourth, it is clear that metaphors can be harmful as well as helpful since they may well lead users to import knowledge not supported by the interface. Our point is not that metaphors have not useful but that as the primary design strategy they may well restrict the range of interfaces designers consider and impose less effective trade-offs than designers might come to if they were to consider a larger space of possible interfaces.

There are, of course, also costs associated in following a physics-based design strategy. One cost is that designers can no longer rely as heavily on users' familiarity with the metaphorical reference (at least at the level of traditional objects and functionality) and so, physics-based designs may take longer to learn. However, the power of metaphor comes early in usage and is rapidly superseded by the power of actual experience. One might want to focus on easily discoverable physics. As is the case with metaphors, all physics are not created equally discoverable or equally fitted to the requirements of human cognition. Still, the new dynamic representations we are now starting to explore as part of an informational physics strategy may well serve as the metaphorical basis for future interfaces.

## 2.9 Acknowledgments

Our views of information visualization and an informational physics approach to interface design have benefited from many discussions with the members of our research groups over the years. We particularly acknowledge Scott Stornetta, Will Hill, Ken Perlin, Larry Stead, Kent Wittenburg, Jon Meyer, Mark Rosenstein, Dave Wroblewski, and Tim McCandless.

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# Chapter 3

# Mental Models and User Models

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## 3.1 Introduction

The expectations a user has about a computer's behavior come from *mental models* (Figure 1); while the "expectations" a computer has of a user come from *user models* (Figure 2). The two types of models are similar in that they produce expectations one "intelligent agent" (the user or the computer) has of another. The fundamental distinction between them is that mental models are inside the head while user models occur inside a computer. Thus, mental models can be modified only indirectly by training while user models can be examined and manipulated directly.

## 3.2 Mental Models

Models are approximations to objects or processes which maintain some essential aspects of the original. In cognitive psychology, mental models are usually considered to be the ways in which people model *processes*. The emphasis on process distinguishes mental models from other types of cognitive organizers such as schemas. Models of processes may be thought of as simple machines or transducers which combine or transform inputs to produce outputs. While some discussions about mental models focus on the representation, the approach here considers mental models as the combination of a representation and the mechanisms associated with those representations (see Anderson, 1983).

A mental model synthesizes several steps of a process and organizes them as a unit. A mental model does not have to represent all of the steps which compose the actual process (e.g., the model of a computer program or a detailed account of the computer's transistors). Indeed, mental models may be incomplete and may even be internally inconsistent. The representation in a mental model is, obviously, not the same as the real-world processes it is modeling. The mental models may be termed *analogs* of real-world processes because they incorporate some, but not all, aspects of the real-world process (Gentner and Gentner, 1983). Mental models are also termed *user's modes* (Norman, 1983) although the expression is avoided here because of confusion with the term "user models" (Section 3.3).

Because they are not directly observable, several different types of evidence have been used to infer the characteristics of mental models:

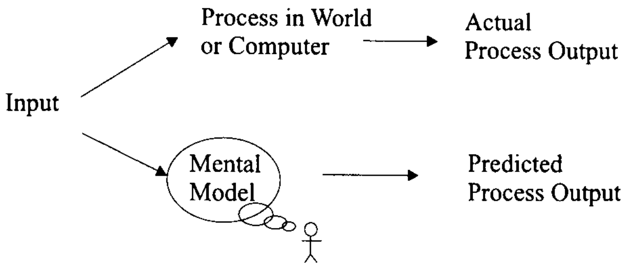


*Figure 1. Process and mental model of that process.*

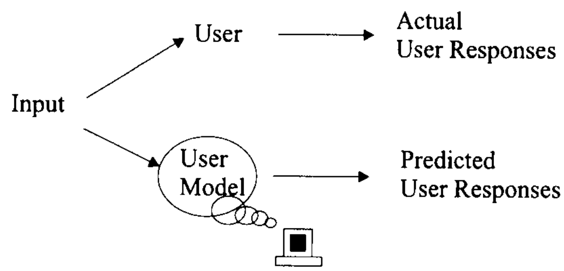


*Figure 2. User responses and user model predicting those responses.*

- **Predictions:** Users can predict what will happen next in a sequential process and how changes in one part of the system will be reflected in other parts of the system. However the most informative aspect of their predictions are often the errors from the model.

- **Explanations and Diagnosis:** Explanations about the causes of an event and diagnoses of the reasons for malfunctions reflect mental models.

- **Training:** People who are trained to perform tasks with a coherent account (i.e., a conceptual model, see below) of those tasks complete them better than people who are not trained with the model.

- **Other:** Evidence is also obtained from reaction times for eye movements and answering questions about processes.

### 3.2.1 Conceptual Models

Because mental models are in the user's head, it is helpful to have models of mental models in order to discuss them. These models of mental models may be termed *conceptual models*. Several classes of conceptual models may be identified:

**Metaphor:** Metaphor uses of the similarity of one process with which a person is familiar to teach that person about a different process (Carroll, 1988). For instance, a filing cabinet for paper records may be used to explain a computer file system. Indeed, the metaphor may be built into the interface (as in the use of filing cabinet icons). The ways in which a metaphor is incorporated into a mental model are difficult to examine and probably vary greatly from user to user. Moreover, a metaphor can be counterproductive because the metaphor is rarely a perfect match to the actual process and incorrect generalizations from the metaphor can result in poor performance on the task (Halasz, 1983). For instance, if word processing is introduced by analogy to typing on paper pages, word-wrapping on the screen makes little sense.

**Surrogates:** Surrogates are descriptions of the mechanisms underlying the process. For a pocket calculator, surrogate models would describe its function in terms of registers and stacks. As noted by Young (1983), surrogate models are not well suited to describing user-level interaction. For example, they provide poor explanations for the learnability of a system (e.g., how to use a calculator to do simple arithmetic operations).

**Mappings, Task-Action Grammars, and Plans:** Another class of conceptual model describes the links between the task the users must complete and the actions required to complete those tasks. Several models have this property. Young (1983) describes task-action *mappings*, which are simple pairings between tasks and actions. He compares this with the surrogate model (above) for describing the performance on simple tasks with three calculator architectures. In distinction to surrogates, he claims that mappings are suitable for describing learnability and as a basis for design.

*Grammars* are of interest because of their ability to describe systematic variations of complex sequences. Specifically, grammars computer dialogs are often described as a type of linguistic interaction. Grammars can be designed which specify actions as terminal nodes. TAG (Task-Action Grammars, Payne and Greene, 1986) is a framework for describing the grammars of specific command languages. It consists of definitions of tasks, categorization of tasks, and rules with schemas for combining tasks.

*Planning* models can also integrate tasks and actions. The GOMS model (Goals, Operators, Methods, and Selection; Card et al., 1983; Kieras, this volume) is a plan-based model which has been used widely to describe complex computer-based tasks.

**Propositional Knowledge:** The process assumptions of conceptual models not always explicit. Johnson-Laird (1983) has proposed that *propositional knowledge* (which he terms a "mental model") is the basis for most logical reasoning. However, his thesis has little direct application to human-computer interaction and it is not considered further here.

### 3.2.2 Mental Models for Computing Systems and Applications

Although mental models have been studied in physics and mathematics, the vast majority of research on them has been based on computer-human interaction. Many aspects of human-interaction with computers involve complex processes, thus people who interact with computer systems must have some type of mental model of those processes. There are several levels of processes in computing about which a person might build a model; these include a computer model such as the hardware, the operating system, software, and applications (such as text editors and spreadsheets). Indeed, an unresolved theoretical issue is whether it is possible to have multiple mental models active simultaneously and, if so, how they might interact.

**Mental Models and Computing Systems:** Computing systems are complex and their use and maintenance requires elaborate planning. Tasks such as system or network administration involve understanding many different subsystems. Users of computer systems also have mental models of those systems, although presumably these are very simple models compared to the system administrators. For instance, a user might have a (largely incorrect) mental model that Internet response times are based on the physical distance email messages have to travel. As noted above, training based on conceptual models about processes can lead to improved performance on tasks requiring an understanding of those processes.

**Mental Models and Computing Applications:** Users of computer applications have mental models of the effects of commands in operating these computing applications. For instance, users have expectations about the effects of entering values in using a spreadsheet. Halasz and Moran (1983) compared the effects of styles of training about simple calculators. Students given task-focused training were better at simple tasks than students trained with conceptual models of the calculator. On the other hand, the students trained with conceptual models were better at tasks that went beyond the initial training.

Borgman (1986) compared two styles of training on a command-based information retrieval system. In one type of training a conceptual model (what she called a "mental model") was used an analogy between the retrieval system and a traditional card catalog. The other training style was giving examples of how specific procedures would be accomplished. As in Halasz' study, the users trained by analogy performed better on tasks that required inferences beyond what was covered in the training.

When a user attempts to apply knowledge from a mental model for one task to another task, the *transfer* may show synergy or conflict. For instance, Douglas and Moran (1983) report that users familiar with traditional typewriters had more difficulty learning about electronic word processors than others. Mental models of "experts" may actually be counterproductive during transfer if they are not relevant to the task at hand. Singley and Anderson (1985) compared transfer among multiple text editors. The transfer was modeled by ACT (Anderson, 1983) in terms of the overlap in the number of rules needed to complete tasks with each of the text editors.

**Designer's Mental Models:** Design of complex systems requires mental models (Simon, 1969). Computer-related design tasks (as well as related design tasks such as the design of video-games, educational applications, and CD-ROMs) may involve the interaction of several different mental models. These may include models of the capabilities of the tools, models of the partially completed work and models of the user's interests and capabilities (Fischer, 1991).

Effective programmers seems to have a conceptual model of their programs as a machine for transforming inputs to outputs. Littman et al. (1986) compared two groups of programmers in a program debugging task. One group was instructed to attempt to systematically understand the program while fixing bugs. The other group was instructed to fix the bugs without attempting to form an overview of the programmer's function. The members of the first group were much better at understanding the interaction of the components of the program and they were also better at fixing the bugs.

### 3.2.3 Structuring Content to Improve Mental Models

The most important practical application of understanding students' mental models is for training. This section explores the issues in enhancing mental models of students with complex training programs. Section 3.3.11 examines interactive student models in which the students' knowledge is modeled to improve the performance, tutoring, and help systems.

Selection of appropriate text and graphics can aid the development of mental models. For instance, parts of a document could be highlighted to emphasize their relation to a particular concept (e.g., Kobsa et al., 1994). Training material about dynamic processes may include diagrams and other techniques for improving the learner's mental models. Hegarty and Just (1993) and Kieras (1988) have examined the optimal level of realism to present in schematic diagrams.

Beyond the local effects of media on mental models, the organization of the entire content of a manual or a course may be designed to improve the development of the user's mental models. *Scaffolding* is the process of training a student on core concepts and then gradually expanded the training. The "training wheels" approach (Carroll, 1990) is a type of scaffolding for training about computer systems.

Animation of data or scenarios which evolve over time should be especially useful for developing mental models because the causal relations in a process can be clearly illustrated. Gonzalez (1996) examined many properties of animations and found that factors such as the smoothness of the transitions were important for performance of tasks which had been presented with the animations. Because performance improved, it may be assumed that the mental models are also improved.

Interaction with a virtual environment can allow users to focus on those topics with which they are least familiar. The utility of organizers for improving the recall of information is well established. Lokuge et al. (1996) used this broad sense of mental models to develop a representation of the relationships among tourist sights in Boston. They then built a hypertext presentation which aids in the presentation of those relationships to people unfamiliar with Boston.

## 3.3 User Models

As indicated in Figure 2, the user model is the model computer software has of the user. In the following sections, several issues for user modeling are discussed (Sections 3.3.1-3.3.6) and then application areas are examined (Sections 3.3.7-3.3.11).

### 3.3.1 User Model Inputs

User models have parameters which can distinguish users. Sometimes these are set explicitly by the user and sometimes they are inferred by the computer from the user's past responses and behavior.

**Explicit Profiles:** In some user modeling techniques, users must create a profile of their interests. For example, in the information filtering technique know as *Selective dissemination of information* (SDI, Section 3.3.8), users must specify what terms match their interests. However, users may not have a clear memory of preferences or may not want to give an honest response. In addition, performance will be better if the user understands the model enough to select the discriminative terms (i.e., has a mental model of the user model mechanism). To some degree all entries in an explicit user profile are a type of self-categorization.

**Inferences from User Behavior:** An unobtrusive recording of movie preferences might simply collect information from a set-top box what movies a person had the set tuned to and how long the set stayed turned to that movie. In addition, assumptions are necessary to interpret this type of data. It is not safe to assume that a user is looking at their TV screen all the time, thus the amount of time a video is displayed on that screen may not be an accurate measure of the person's interest in that material. On the other hand, this type of data often has considerable value. Morita and Shinoda (1994) found a positive correlation between the amount of time a person spent look at a document and the ratings of interest in that document.

### 3.3.2 Degree of Personalization

User Models may be personalized to different degrees. They may range from baserate predictions to totally individualized predictions. All of these models should be better than random predictions.

**Baserate Predictions:** A *baserate* prediction is what would be expected for any individual (Allen, 1990). Baserate models might not even be considered user models since they are not differentiated across individuals. One example is of a baserate predication could follow statistical norms (e.g., that a person would like a

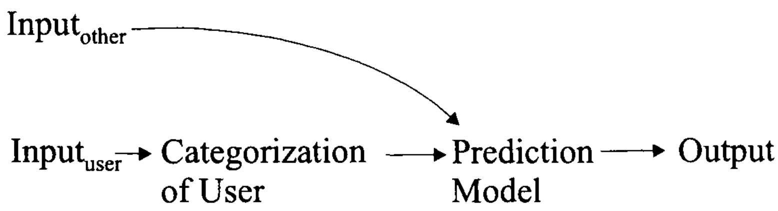


*Figure 3. Categorization of user input.*

best-selling book). Other types of baserate prediction could be derived from laws or social norms. In some cases, when the population is very consistent, baserates may be very accurate. In other cases, they may apply to only a small portion of the population.

**User Categorization:** Rather than developing separate models for each individual, users are often categorized and models developed based on those categories (as illustrated in Figure 3). Of course, the complexity of the categories may vary greatly from simple demographics (e.g., age) to complex personality constructs. There are many examples of categorization in user modeling. One example is the classification of utterances according to their function as *speech acts* (Section 3.3.10).

Another example of a user categorization is the novice-expert distinction. It might be expected that there would be consistent differences in the behavior of experts and novices which could be useful, for instance, in constructing different types of training. While there are clearly differences in the knowledge of novices and experts on a given task (e.g., Egan, 1988; Mayer, 1988), difficulties arise when attempting to classify people as novices or experts or in attempting to generalize expertise in one area to expertise in other areas. There are many dimensions of expertise and although users may be an expert on one set of commands, they may be novices in other areas. Thus, broad classification of users as experts or novices does not often seem to be helpful.

Still another example of categorization of users is a *stereotype* (Rich, 1989). Rich's "Grundy" system predicted preferences for fiction books, primarily using a user's self-descriptions (e.g., feminist) with adjectives as inputs. Unfortunately, it was difficult to tell whether Grundy's predictions were better than simple baserate predictions. To provide control conditions for a Grundy-like system for predicting book preferences, baserate predictions were found to make significantly better predictions than a random selection and a simple 'male/female' dichotomy further improved the predictions (Allen, 1990).

Kass and Finin (1991) describe GUMS (Generalized User Modeling System) in which they introduce the notion of hierarchical stereotypes and inference mechanisms based on that those stereotypes. This formalism is helpful for reasoning about users; however, it seems removed from the usual psychological notion of a stereotype. For instance, GUMS stereotypes include 'rational agents' and 'cooperative agents'. In addition, it is not clear that the stereotypes people have can be combined with or derived from other stereotypes in a systematic way.

### 3.3.3 User Models in Complex Systems

**Blackboard Systems:** A user model may be a component of more a complex system which includes other components. In some cases, a user model is said to contain all the knowledge a system has about the user. For instance, Wahlster and Kobsa (1989, p. 6) state: "A user model is a knowledge source in a NL dialog system which contains explicit assumptions on all aspects of the user that may be relevant to the dialog behavior of the system". In other cases, that knowledge is subdivided into several specific models such as task models and situation models. In training systems (Section 3.3.11), the "student model" holds task-relevant state and the "user model" applies to long-term knowledge about the user such as demographic information. This situation is similar to the suggestion that several different mental models may be active for a programming task (Section 3.2.2).

Figure 4 shows a typical collection of knowledge sources which includes user, task, and situation. The inputs are combined with data from various repositories on a blackboard. As described in the previous section, the *task expert* has information about what the user is trying to accomplish and possible strategies for accomplishing those goals. The *situation expert* contributes knowledge about the environment in which the

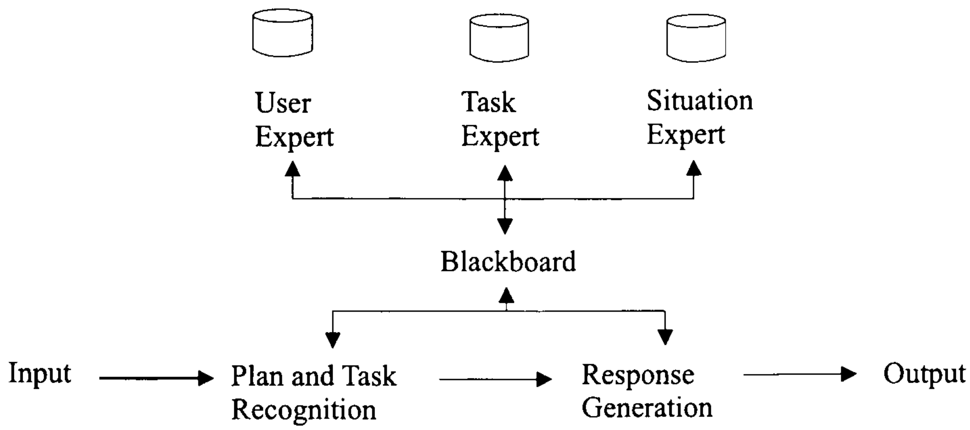


*Figure 4. Typical components of a blackboard system.*

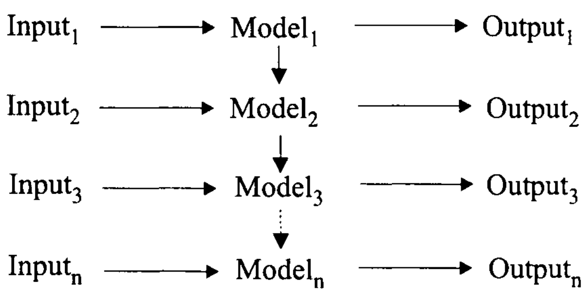


*Figure 5. Adaptation of model across sequential events.*

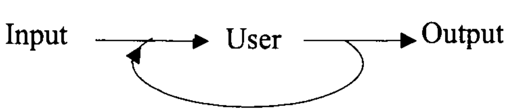


*Figure 6. Model with feedback.*

user is trying to complete the task. There are many possible ways of organizing the knowledge required for this type of system. Other components which have been proposed include *system, domain,* and *discourse* processors. Clearly, it is possible to partition these models in many different ways and simply proposing different sets of models is not necessarily helpful. Modularity is generally a useful design principle. Unfortunately, information about users may be very tangled and inconsistent. It may be difficult to separate the user model from the other models components. Even for Figure 3, it seems somewhat arbitrary to declare that the user model is just the categorization stage and does not include the Prediction Model.

**User Agents:** An *agent* may be considered as just one module of a complex system (e.g., one of the experts in Figure 4). However, a more interesting sense of the term "agent" is as modular system which acts on behalf of the user. A user agent might simply provide information about the user. For instance, it might search a database of the user's writings to answer a question on behalf of the user. An active user agent may negotiate on behalf of the user. This mean that user agents must have a model of the relative values of alternative outcomes for the user.

### 3.3.4 Adaptive Models, Machine Learning, and Feedback

User models are often said to *adapt* to users. However, there are different senses in which a model may be adaptive. In the simplest sense, a model is adaptive if it gives different responses to different categories of users. A more interesting sense is that a model adapts as it gains experience with an individual user. Adaptation may be within connected sequences of events or across different events. Figure 5 illustrates a model which keeps state across a sequences of inputs such as a conversation, information retrieval session, or task.

As indicated in Figure 6, *feedback* uses output from the model to refine it. This would be most common for models in which adjustments are made across a sequence of events for future trials. Most often feedback is an automatic process, however, having users refine their profiles could also be considered a type of feedback. The idea of feedback originated with control theory in which only a few parameters in the model would typically change; however, in computer models, the structure (i.e., the program) of the model itself may change.

User models which change and improve their performance as a result of greater experience are said to show *machine learning*. Some examples of machine learning algorithms are neural networks, genetic algorithms, clustering, and case-based reasoning. An important distinction is whether all training examples are maintained by the model (as in case-based reasoning) or whether other data are compressed to form a *representation* of the original training set.

### 3.3.5 Tasks and Planning

User models are often distinguish according to whether they apply to "short-term" or "long- term" interactions, however, this blurs several issues of adaptation and feedback. Typically, short-term models describe the user performance on specific tasks.

**Tasks and Task Models:** The task in which a person is engaged and plausible strategies for completing it greatly constrain the behavior of that person. Indeed, the task in which a person is engaged is often more important than individual differences for predicting user behavior (see Mischel, 1968). The interaction of tasks with behavior is so strong that there is a tendency to identify tasks or even to define new tasks to explain ambiguous behaviors. On the other hand, there are many cases in which individual differences have a major effect (Egan, 1988).

A *task model* (Sleeman, 1982) is a description of a task as well as strategies for completing that task. In some cases, the task model may include incorrect methods for completing the task. In situations when tasks are clearly defined, mental models and user models reflect the task structure. For instance, student models (Section 3.3.11) are typically focused on a student completing specific tasks.

**Planning and Plan Recognition:** Planning often proceeds as a cycle of determining goals or subgoals and finding methods for completing those goals. GOMS (Section 3.2.1; Kieras, this volume) is a planning model of how people complete command sequences. There are several different types of planning mechanisms and the nature of planning has been widely discussed recently. Some important dimensions for planning are whether the plans are fixed once they have been established or whether they may be modified depending on the situation.

When users complete a task they, presumably, have strategy for what they are doing. However, a human being or computer with whom they are interacting many not know what strategy the users have or even what task they are trying to accomplish. The process of *plan recognition* would involve identifying the task and the strategy. Although Figure 5 illustrated a user model in which the model is known, the figure might also be applied to plan recognition in which an observer must infer the model (i.e., the plan) and maybe the inputs (i.e., the task) given the outputs and a few contextual clues. For instance, a person who comes to a train ticket window is likely to be asking for information about trains but could, instead, be asking for other types of information. As suggested by J. Allen (1983, p. 108), "[People] are often capable of inferring the plans of other agents that perform some action."

A wide variety of techniques has been proposed for plan recognition. Most often some type of grammar is fit to responses although statistical methods such as neural networks could also be applied. Plan recognition may benefit from information about the user. For instance, knowing accents and idiosyncratic expressions may be useful for speech utterance understanding systems. The closely related problem of interpreting a student's strategy in solving a problem is an essential component of intelligent tutoring systems (Section 3.3.11).

Another tradition of inferring information about people from the context of their actions is *attribution theory* (Kelley, 1963). This describes the processes by which people make inferences about internal state of other people. For instance, *social norms* might be employed in assessing how a person might be expected to be behave in a given situation. A person who behaves differently from the norm is likely to be judged as showing an *intention* for that action.

### 3.3.6 Evaluation of User Models

**Evaluation Criteria:** The main criterion for the *effectiveness* of a user model is in predicting important behavior which facilitates the user's activities. Among the components contributing to this are:

- *Relevance* requires that models make predictions that apply to the target behavior or user goals.
- *Accuracy* requires that the models make correct predictions for at least one task and situation.
- *Generality* of the model requires robustness despite changes in tasks, situations, and users. In addition, the model should be *scaleable* with increased numbers of tasks, situations and users. Too many existing systems have been tested only on "toy" data sets.

- *Adaptability* of the model requires that it not be *brittle* in response to changes in user behavior.

- *Ease of development and maintenance* is whether the effort in maintaining the user model is worthwhile for the user. A typical factor is whether the inputs are collected unobtrusively. For instance, if the model has to be maintained by an individual, is it clear to the user how to do that?

- *Utility*: The model should improve the user's behavior. For instance, adding new commands to an interface based on a models might confuse the user.

**Evaluation Techniques:** While the ultimate test of a user model is how well it satisfies the criteria listed above, it is often difficult to test a model as a whole in field conditions. However, to test parts of it in limited environments. For a given set of inputs it is worth choosing the model which give the best predictions. Another standard for measuring the performance of a user model is to compare it to the performance of a human being who is given the same inputs. This is a control condition for the model. It may be called a *Turing control* condition because it is a limited type of Turing Test (Allen, 1990).

The extensive literature in psychology on techniques for assessing individual differences and personality types can be applied to the evaluation of user models. Standard assessment criteria such as *reliability* and *validity* can be adopted. Reliability is measured as the consistency by which different samples of a person's behavior are classified the same. Validity is the agreement of a model's classification with evidence from other aspects of behavior. *Incremental validity* (Mischel, 1968) is the use of the simplest set of independent variables and inference mechanisms to generate classifications.

**Social Implications of User Models:** Beyond the evaluation of the performance of user models, they may be evaluated in terms of their effect on individuals in society and on society itself.

*Privacy* may be compromised by having large amounts of detailed information about individuals stored on-line. Private information may be breached and used in unauthorized ways. An alternative is not to store sensitive information about a person but only categorical information and demographics. Another possibility is to have the most personal details of these systems stored locally and under user's control. For instance, an agent acting on behalf of the user could have access to the data and respond to inquires.

If agents (Section 3.3.3) make decisions on behalf of human beings based on user models, then the users may be removed from those actions and may not feel *responsible* for the actions of their agents. Eventually laws are likely to be developed to clarify the extent of responsibility a person has for agents acting on their behalf.

Effective user modeling may greatly affect the behavior of individuals and their *relationship to society*. For instance, information services can be tailored for individuals. Paradoxically, however, the ability to personalize highlights the similarity among people.

### 3.3.7 User Models for Supporting Completion of Routine Tasks

Streamlining Routine Command Sequences: It is widely recognized that the completion of tasks in HCI follows regular patterns. Several proposals have been made for systems which learn patterns of computer commands by purely statistical regulation, or inference underlying tasks (e.g., Hanson et al., 1984). From a different approach, Desmarais et al. (1991) apply plan recognition to parsing user actions in order to recognize inefficient user behavior and to suggest shortcuts. "Eager" is a programming-by-example system (Cypher, 1991) which attempted to anticipate and support the placement of widgets by GUI designers by observing patterns of their responses.

**Agents and Routine Office Tasks:** User agents have been applied to routine office tasks such as scheduling appointments on an electronic calendar and prioritizing email. Typically, these agents do not model goals and complex chains of events. Because the tasks being modeled are repetitive, there is sufficient data to apply machine learning to find salient attributes. The agents also often adapt across users and across tasks for a given user.

Dent et al. (1992) describe a learning apprentice and apply it to personal calendar management. The system uses decision trees to characterize previous calendar scheduling. For example, it created a rule that theory seminars were held on Mondays. After training, the model performed slightly better than rules hand coded by the researchers. Maes and Kozerick (1994) describe two interfaces with agents: an email assistant and a calendar scheduling assistant. The agents predicted meeting schedules with an inference mechanism known as case-based reasoning. Small improvements in the confidence level reported by the agents' when their predictions when those predictions were correct.

### 3.3.8 User Models for Information Access

**Information Retrieval and Information Filtering:** The most widely studied approach to information needs is *information retrieval* (IR) (Dumais, 1988). The usual paradigm in information retrieval is to find documents which match a user query. This requires representations of the document collection and the query as well as algorithms for comparing the two representations. The algorithm for comparing queries to documentation may be simple keyword matches or complex derivation of verbal associations. In terms of user models, the IR algorithms integrate input from users and produces a set of relevant documents which are, ideally, those the users would have selected for themselves.

An IR model may be tuned to improve its selections by *relevance feedback*. The user identifies the most relevant documents and terms from those are used as inputs for a revised model (see Figure 6) to use in retrieving documents from the document representation. While relevance feedback is usually thought of as refining a query this may also be seen as developing a user model (Figure 5). For instance, Foltz and Dumais (1992) represented users' interests as a points in latent-semantic indexing (LSI) space.

Although there are parallels between IR and generalized user modeling, it seems that user modeling could be used even more in IR systems. The IR task is often implicit (e.g., a research topic the user is exploring). Demographic information about the user has, typically, been little used in the IR literature (Korfhage, 1984). There have been suggestions for long-term user models as part of IR systems. For instance, a user's interest across several sessions could be recorded and used to improve responses to queries but little work has been done.

*Information filtering* is a variation of IR; specifically, it is a type of text categorization. Filters identify new information which matches long-term, relatively stable interests. Selective dissemination of information (SDI, Luhn, 1958) usually employs an explicit profile of keywords. In some systems, these word lists are maintained by the users and it not clear if people are good at maintaining these profiles. Feedback in SDI is accomplished by the users making changes in their profiles.

Agents (see also Sections 3.3.3 and 3.3.7) were applied to adaptive information filtering by Sheth (1994). Specifically, genetic algorithms were employed as a highly adaptive model of NetNews preferences. This agent was able to acquire initial models for several subjects. It was also able to adapt to changes in simulated preferences ratings.

**Hypertext Linking:** Hypertext allows users to browse text by linking information objects together. Links may be selected that would be relevant to a specific task of use or interest to a given user. One example of adaptive hypertext is from KN-AHS (Kobsa et al., 1994) in which, for instance, the fact that a user checks a term in a glossary is used to infer that the user is not familiar with that term and a full explanation of it is included if it is used in later text. Thus, tailoring links in a hypertext may be related to community ratings on web pages (Section 3.3.8), to language generation (Section 3.3.10) and interfaces for training (Section 3.3.11).

Graphical views of objects and links which have been frequently accessed have been described as "history-enriched digital objects" (Hill et al., 1993, also see Lesk and Geller, 1983). Typically, these show total counts (i.e., baserates).

On the Web, there is often a delay in accessing information, and it is helpful to predict what topics the user will browse next. The anticipated material can be *prefetched* so it will be available locally if the user decides to access it. Clearly, prefetching would be more efficient with accurate models of what the user is likely to access.

**Information and Aesthetic Preferences:** Information preferences may include movies a person would like to view or books or news articles they would like to read. Information and aesthetic preferences may be distinguished from information needs (previous sections) in not being associated with a particular task. Because the are based on easily-described tasks, it is difficult to model users' processes involved in aesthetic preferences.

Probably the most successful technique for predicting information and aesthetic preferences is known as *collaborative filtering* (Section 3.3.9) although other techniques for predicting information preferences have been examined. These include the stereotypes of the "Grundy" system described earlier (Section 3.3.2).
A content-based system which modeled music preferences was proposed by Loeb (1992). Songs were categorized and models developed for each category of song based on ratings made by the user just after the song had finished. This is, essentially, an IR relevance feedback mechanism.

***Table 1. Hypothetical ratings of seven items by four users.***

| | 1 | 2 | 3 | 4 | 5 | 6 | Target |
|---|---|---|---|---|---|---|---|
| **1** | 9 | 1 | 4 | 8 | 3 | 0 | 2 |
| **2** | 3 | 0 | 9 | 2 | 3 | 8 | 1 |
| **3** | 2 | 8 | 7 | 9 | 3 | 1 | 7 |
| **4** | 8 | 3 | 3 | 7 | 8 | 2 | ? |

***Table 2. Correlations of ratings among four users from Table 1.***

| | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| **1** | - | | | |
| **2** | -0.14 | - | | |
| **3** | +0.10 | -0.40 | - | |
| **4** | +0.74 | -0.49 | -0.11 | - |

Yet another proposal for predicting information preferences is based on "interestingness" (e.g., Hidi and Baird, 1986). This is, essentially, a baserate prediction and might work if there is consensus over what is interesting. However, where there are substantial individual differences, the problem of determining what is interesting seems no easier than the original task of predicting preferences.

### 3.3.9 Collaborative Filtering for Information Preferences

While most user models attempt to model the user's thought process, other mechanisms are possible. Recently, statistical mechanisms based on identifying individuals with similar preferences have been proposed (Allen, 1990). With collaborative filtering (also known as "community ratings" or "social learning"), there is no attempt to model the underlying cognitive process. Rather, the preferences of other individuals are employed, the other individuals integrate the inputs.

**Example:** While users may differ greatly in their preference ratings for different items, subgroups of users may be quite similar to each other. Table 1 shows hypothetical ratings by four users for seven documents. Table 2 shows the correlation coefficients of the ratings on a 10-point scale for the users. The ratings of Users 1 and 4 are similar. Thus, User 1's ratings could be a predictor for User 4's ratings (specifically, predicting a rating of 2 for the target item). Beyond the simple correlations, multivariate statistics, such as *multiple linear regression,* combine evidence from several different users into a single predictive model. For the data in Table 1, a multiple linear regression gives an $\mathbf{R}^2 = +0.77$ which is better than any of the simple correlations for that user in Table 2.

**News, Videos, Movies, and Music:** Allen (1990) used collaborative filtering to make predictions about preferences for news articles, however, few of those correlations were substantial. This may have been because of the small number of participants or because news preferences are not stable and are difficult to predict.

Collaborative filtering has been more successful predicting preferences for movies. Hill et al. (1995) report studies of video preference predictions with data collected from the Internet. 291 respondents made 55,000 ratings of 1700 movies. Three of four video recommendations made by the service were highly evaluated by the users of the system. In Figure 7, individualized predictions show a much stronger correlation with the user's preferences than do the predictions of the movie critics.

Collaborative filtering has also been applied to recommend music preferences. Shardanad and Maes (1995) developed the "Ringo" system (also known as "Firefly") in which ratings were collected from users by email. Modified pairwise correlation coefficients were calculated for 1000 of the people who made ratings. Predictions of preferences based on these correlations were somewhat better than the baserate predictions.

**Limitations and Extensions of Collaborative Filtering:** A number of constraints determine whether collaborative filtering will be effective in real-world applications. Ratings are often used to build the filtering models. Ratings are not direct measures of the target behavior and may be affected by many incidental factors. The underlying preferences must also be fairly stable. For instance, musical tastes may change depending on time of day, mood, and what is going on around a person. The ratings a person provides must not change very much while predictions are being made. If the community is heterogeneous and if there is a wide variety of opinions about the material, a relatively large number of other people must make ratings.

Beyond the basic correlational techniques described above, a number of extensions for collaborative filtering techniques may be considered. While collabo-

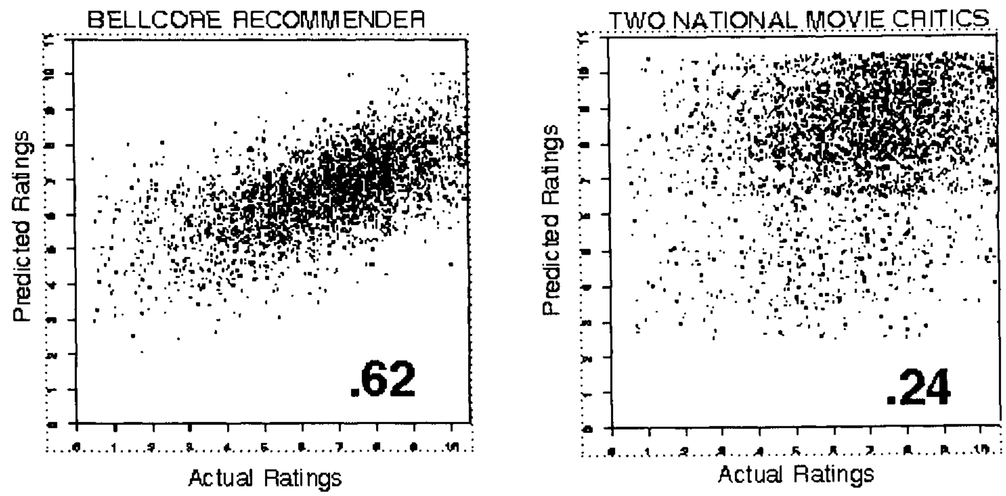


*Figure 7. Video recommender scatterplots.*

rative filtering is an effective technique, it seems likely that predictions with a mixture of community ratings and a limited amount of content (categorization) could be better still. Preferences in one domain (e.g., movies) could be used to make predictions across domains (e.g., books). Indeed, a generalized profile might be developed in which a wide range of a user's preferences were combined and compared with other people's preferences. In addition to matching a person with an information object, user models can also be used to match one person with another. For instance, people with similar interests might be identified based on correlations their aesthetic preference ratings. In some cases, information about users may be used to *initiate* the presentation of information to them. This would often be *targeted advertising* (i.e., personalized commercial information); however, other types of information such as health messages and community information could also be triggered.

Predictions may be made by friends of the user (Allen, 1990). In the previous section the subjects in the control condition had relatively little information about the preferences of the person for whom the predictions were to be made. Collaborative filtering might also be used to make predictions about the preferences of a group such as a family or friends attempting to find a movie which they would enjoy. If the group provides ratings as a unit, the predictions should be similar to those made for individuals. Alternatively, group prediction could be synthesized from predictions for the individuals in that group. In that case, a simple technique would be to find the item which was highest, on the average, across users. However, group dynamics such as the dominance among members of the group would be difficult to model.

### 3.3.10 User Models and Natural Language

Natural language processing is complex and involves many user modeling issues. This section is divided into discussion of low-level modeling issues in the speech chain and higher-level issues affecting understanding and generation. Related issues are also considered in the following section on training (Section 3.3.11) and in Section 3.3.5 under subsection on plan recognition.

**Speech Chain:** The activities associated with the production of speech may be said to form a *speech chain*. There are individual differences at many steps in the speech chain. These include speaker characteristics in speech recognition, word senses, complexity of vocabulary and grammar, and dialect. These differences apply to both language understanding and generation.

Speech recognition attempts to identify the words in user utterances. Because individuals differ substantially from each other in their speech, it is helpful to have a model of their characteristic speech patterns. These individualized models, known as *speaker-dependent* models, can be used to transform responses for further processing. However, a detailed review of the literature on user models for speech recognition is beyond the scope of this paper.

Neural networks were used classify gestures of a person wearing a data glove and then to produce speech based on those gestures (Fels and Hinton, 1995). The neural networks adapted to the idiosyncratic responses of the user. After 100 hours of training, a person gesturing with the data glove was able to use the system to produce speech somewhat better than a speech synthesizer.

**Language Generation and Understanding:** Communication between people is much more than the transmission of literal messages which can be easily deciphered with a dictionary and simple parser. Rather, communication is highly context and task dependent and both sender and receiver have complex expectations about the for the interaction. These expectations are described as *conversational maxims* (Grice, 1980). They are: Quality (truthful and accurate information), Quantity (neither more nor less information than is required), Relation (information appropriate to the task), and Manner (clear and unambiguous information).

The emphasis on the *function* of messages instead of on their literal meaning has led Austin (1962) to categorize them in terms of *speech acts*. Speech acts may be *indirect* (Searle, 1980) as in irony where is intended message is the opposite of the literal message. Purposeful conversations have a regular structure or pattern of speech acts (Winograd and Flores, 1986). For instance, we would expect a offer of help to be followed by an acceptance or rejection of that offer.

In order to engage in communication human beings may be said to have mental models of the task, the context, and the other participants. If the communication is between a human being and a computer (whether as a task-oriented dialog or as natural language dialog) then the computer may be said to have a user model of the human being. In order to fulfill conversational maxims and to initiate speech acts, in language generation, a speaker or agent must consider long-term characteristics of the listeners such as their knowledge level. In addition, the speaker may monitor the reactions of the listeners during the conversation to ensure the intended communication is received. Likewise, the listener could expect that the communicator was following conversational maxims. Recognition of speech acts by a listener may be viewed as a type of plan recognition (see Section 3.3.5). As noted earlier, plan recognition would include many aspects of user models such as the situation, the speaker's appearance, and previous experiences with the speaker.

### 3.3.11 User Models for Tutoring, Training, and Help Systems

Instructional interaction between a computer and a human being may be viewed as a specialized conversation. For workers using computers to complete routine tasks, the actions in which the user may be engaged are generally highly constrained. Thus, for help systems which try to give advice to those workers, little inference is about the task is needed. However, the task and the user's interpretation of it are generally less well known for training or tutoring contexts than for help system. Thus, plan recognition is an important part of training and tutoring systems.

Tutoring and training are closely related to each other although training is focused on teaching skills needed for a particular task while tutoring is applied to learning general skills such as reading and mathematics. Help systems support users who are attempting to complete tasks with a computer system. These systems differ in their style of interaction with the student. For instance, the system may interrupt the user in the middle of a task to provide advice, or it may wait for the user to request information. Similarly, the system might follow a prescribed set of tasks or it might schedule tasks for the student depending on the model of the student's knowledge.

Personalization in tutoring may be modeled by observing the conversations between tutors and students. There have been several studies of how human tutors adapt their interests to a given student and to assess the human tutors' strategies for modeling the user (e.g., Grosz 1983; Stevens, et al., 1982).

The models of users or students in training, tutoring, and help systems combine aspects of mental models, user models, and task models. *Student models* usually describe the student on a given task but generally do not include demographic information about the student. In the style of Figure 4 separate "user models" with long-term information are often included in these systems.

The key for training systems based on student models is in recognizing inefficiencies and errors in the student's performance and determining where they came from. The simplest approach to recognizing errors is to make a catalog of errors for a given task. However this is not always so easy, there may be several correct ways to solve a task.

A more general approach to recognizing errors is known as *differential modeling* (Burton and Brown, 1979). This compares a student's performance to that of an expert engaged in a similar task. The models a student has of the task may not be like a expert's models of that same task (Clancey and Lestinger, 1981). Because they are novices, students' models are generally simplified or incomplete (see Section 3.3). Indeed, the student's model may have inconsistent and even illogical strategies for completing tasks.

Determining the cause of an error can be very tricky; an error may be caused by a combination of factors. Errors may be modeled as the result of specific